



Bahir Dar University

Bahir Dar Institute of Technology Faculty of Computing

Requirement Analysis Document (RAD)

for Human Resource Management Portal

Submitted to the faculty of computing in partial fulfillment of the requirements for the
degree of Bachelor of Science in Computer Science

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Declaration

The Project is our own and has not been presented for a degree in any other university and all the sources of material used for the project have been duly acknowledged.

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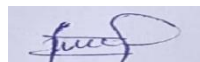
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List of acronyms

Abbreviation	Full Form / Meaning
API	Application Programming Interface
CRC	Class-Responsibility-Collaboration
CRUD	Create, Read, Update, Delete (implied in logic models)
DFD	Data Flow Diagram
HR	Human Resources
HRM	Human Resource Management
HRMS	Human Resource Management System
IDE	Integrated Development Environment
ID	Identification
MFA	Multi-Factor Authentication (mentioned in change cases)
OO	Object-Oriented
REST	Representational State Transfer
RBAC	Role-Based Access Control
UI	User Interface
UML	Unified Modeling Language
UX	User Experience

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Abstract

In modern organizations, efficient management of human resources is essential for improving productivity, ensuring transparency, and supporting informed decision-making. However, traditional HR management processes are often fragmented, manual, and inefficient, leading to data inconsistencies and limited accessibility. This project, **Human Resource Management System (HRMS)**, addresses these challenges by developing a centralized platform for managing employee information, payroll and organizational workflows as well as Clearance levels and health permissions.

The system is built using **React** for the frontend, **Node.js** for the backend, and **Postgre SQL** as the database, providing a scalable, secure, and high-performance solution. The centralized architecture enables real-time access to employee data through a unified and user-friendly interface, allowing HR managers and administrators to efficiently manage workforce operations.

The proposed HRMS incorporates key features such as role-based access control, automated HR processes, performance tracking, and secure data handling. By streamlining HR operations and reducing manual workloads, the system enhances organizational efficiency, improves data accuracy, and supports better strategic planning and decision-making.

Chapter One: Introduction

1.1 Background

Human Resource Management is a key function in large organizations, particularly in higher education institutions where a diverse workforce supports teaching, research, and administrative services. Modern HR systems are used to manage employee records, leave, clearance, and other staff-related processes efficiently. Automating these processes helps organizations reduce delays, improve transparency, and ensure accurate record keeping.

Bahir Dar University is one of the well-established public universities in Ethiopia. It was established in 2000 and is located in Bahir Dar city, in the Amhara Regional State. The university has multiple campuses and academic units, including colleges, institutes, and schools. It employs thousands of academic, administrative, and support staff organized under different departments and management structures. The university is led by a president and vice presidents, with administrative and academic units operating at various levels.

The mission of Bahir Dar University is to produce competent graduates, conduct problem-solving research, and provide community services that contribute to national development. Its vision is to become a leading research-oriented university in Africa, recognized for academic excellence and innovation. The university's core values include quality, integrity, accountability, inclusiveness, and commitment to public service. These principles guide its operations, including the management of human resources.

Bahir Dar University currently uses an internal Human Resource Management System to support HR activities. However, some business processes, particularly employee clearance and leave management, are not fully efficient. Clearance processes during staff transfer, resignation, retirement, or contract completion often require manual approvals and follow-ups across multiple units. Similarly, the leave management process lacks full automation and real-time tracking. This project focuses on improving and automating the clearance and leave management processes within the existing HR system to enhance efficiency, accuracy, and service delivery at Bahir Dar University.

1.2 Statement of the Problem

Human Resource Management at Bahir Dar University is currently facing serious operational challenges due to limitations in the existing HR system. Although the university previously had a digital HR system, it is currently not functional because of hardware failure. As a result, critical HR activities are being handled manually, which has negatively affected efficiency, accuracy, and service delivery. These challenges hinder effective human resource administration and place additional workload on HR staff and university employees.

One major problem is the absence of a functional and automated clearance system. Clearance processes related to staff transfer, resignation, retirement, or contract completion are handled manually through paper-based procedures. This creates delays, makes tracking difficult, and overwhelms HR officers, especially when approvals are required from multiple departments. The lack of a centralized clearance system increases the risk of missing information and incomplete clearance records.

Another significant issue is the reliance on paper-based processes for payroll-related communication. Currently, HR officers must manually prepare and transfer paper documents listing employees who are eligible for payment. This approach wastes time, increases paper consumption, and leads to delays and possible errors in salary processing. It also causes inconvenience for employees who must repeatedly follow up on their payment status.

Furthermore, the downtime of the HR system limits access to real-time employee information and reduces data security. Manual records are vulnerable to loss, damage, and unauthorized access, and there is no effective mechanism for tracking changes or ensuring accountability. These challenges highlight the urgent need to improve and automate the HR system by introducing a reliable clearance system and improving HR processes to reduce paper usage, save time, and enhance efficiency at Bahir Dar University.

1.3 Objectives

The objective of this project is to address the operational challenges in the human resource management practices of Bahir Dar University by developing an improved Human Resource Management System that automates and streamlines key HR processes. The project focuses on improving the clearance

process, leave management, and employee name transfer for payroll processing to enhance efficiency, accuracy, and service delivery.

1.3.1 General Objective

To design and develop a centralized and dependable Human Resource Management System (HRMS) capable of supporting efficient, secure, and accurate management of human resource information, processes, and administrative workflows across the institution.

1.3.2 Specific Objectives

- To analyze existing human resource processes related to employee clearance, leave management, and payroll name transfer in order to identify functional requirements, operational gaps, and opportunities for process improvement.
- To implement an automated employee clearance module that enables digital approval workflows and real-time status tracking across all relevant departments.
- To develop a leave management module that allows employees to submit leave requests electronically and provides supervisors and HR personnel with tools to review, approve, and monitor leave records efficiently.
- To design and implement a digital name transfer mechanism to securely communicate approved payroll-related employee information from the Human Resource office to the Finance department.
- To establish a centralized HR database that securely stores employee records related to clearance, leave, and payroll processes.
- To enforce role-based access control to ensure that only authorized users access, manage, and update HR information in accordance with defined responsibilities.
- To reduce paper-based human resource operations and processing time by automating key workflows, thereby improving accuracy, accountability, and operational efficiency.

1.4 Methodology

1.4.1 Requirement Gathering Methods

To ensure that the Human Resource Management System (HRMS) meets the needs of Bahir Dar University, multiple requirement-gathering methods were employed. Interviews were conducted with HR officers, administrators, and selected employees to understand existing HR workflows, challenges, and expectations from the system. These interviews provided detailed insights into daily HR operations, including clearance processes, leave management, attendance tracking, and payroll preparation.

Document analysis was also conducted on existing HR records, including employee registration forms, attendance sheets, leave requests, and payroll records. This helped to identify inefficiencies, redundancies, and gaps in the current manual or semi-digital HR processes. Direct observation of HR activities further provided first-hand insights into how employee data is recorded, stored, and managed, highlighting operational bottlenecks and delays.

Additionally, a literature review was carried out on HR management best practices, labor regulations in Ethiopia, and modern HR technology trends. This ensured that the system design aligns with legal requirements and international HR standards. The gathered requirements were then structured and prioritized, focusing on automating the clearance process, leave management, and employee name transfer for payroll purposes.

1.4.2 Analysis and Design Methodology

The system is modeled using object-oriented (OO) analysis and design techniques to accurately represent system components and their interactions. Object-oriented modeling is applied to key elements such as employees, user accounts, departments, clearance records, leave requests, and payroll transfers. This methodology ensures modularity, scalability, maintainability, and reusability of system components.

Unified Modeling Language (UML) diagrams, including use-case diagrams, class diagrams, and sequence diagrams, are used to represent system functionality, user interactions, and data flow. Use-case diagrams illustrate how different users interact with the system, class diagrams define the structure and relationships among system entities, and sequence diagrams show the flow of activities in processes such as clearance and payroll preparation. Tools such as Lucidchart and Draw.io are used to

develop these diagrams, while flowchart and data flow diagram tools model HR workflows and data movement across departments.

1.4.3 Implementation Methodology

The HRMS is implemented using modern web technologies to provide a secure, scalable, and user-friendly system. The frontend interface is developed using React, enabling a dynamic and responsive experience for HR staff and employees. The backend is built with Node.js, which handles server-side logic, RESTful APIs, authentication, and database communication.

PostgreSQL serves as the database management system, storing employee records, leave requests, clearance information, and payroll transfer data. Its flexible schema and scalability make it suitable for managing both structured and semi-structured HR data. Visual Studio Code is used as the primary development environment for coding, debugging, testing, and maintaining the system. The selected technology stack ensures maintainability, security, and the ability to accommodate future enhancements.

1.5 Feasibility

Economic Feasibility

The system is economically feasible because it relies on open-source technologies such as React, Node.js, and PostgreSQL, eliminating licensing costs. Hosting can be performed on free-tier cloud services during development. Automation of clearance, leave, and payroll processes will reduce paper usage, administrative workload, and operational inefficiencies, leading to long-term cost savings.

Technical Feasibility

The development team has the necessary skills in web application development, database management, and system integration. The technology stack is widely supported, secure, and scalable. Cloud-based deployment ensures reliable performance and accessibility for users across different university departments.

Time Feasibility

The project is achievable within the allocated academic timeframe. Requirement gathering and system design will be completed in the initial phase, followed by modular development of frontend, backend,

and database components. Sufficient time is allocated for system testing, including functionality, usability, and security checks, ensuring that the HRMS is fully functional and reliable.

1.6 Beneficiaries or Significance of the Project

The development and deployment of the Human Resource Management System (HRMS) at Bahir Dar University will provide significant benefits to various stakeholders, including HR administrators, management, employees, and the institution as a whole.

HR Administrators and Management: The system will improve efficiency by automating critical HR processes such as employee clearance, leave management, and payroll name transfer. HR staff will be able to quickly access accurate and up-to-date employee information, reducing administrative workload and minimizing errors. Centralized HR data will allow management to monitor workforce performance, attendance, and compliance with organizational policies more effectively, facilitating better decision-making and resource planning.

Employees: Staff members will benefit from increased transparency and accessibility. They will be able to view personal information, leave status, clearance updates, and payroll details in real time. Faster processing of HR requests and reduced reliance on paper-based procedures will enhance employee satisfaction, trust, and engagement with HR services.

Bahir Dar University: The institution as a whole will experience operational improvements. Automating manual HR workflows will reduce paper usage, save employee and administrative time, and lower operational costs. The system also ensures compliance with labor laws and internal policies, mitigating risks associated with errors or delays in HR processes. Additionally, the HRMS provides a scalable platform that can accommodate future growth, further automation, and integration with other university systems.

Overall, the project contributes to the modernization of HR practices, improving efficiency, accuracy, and transparency. It not only benefits the immediate users but also sets a foundation for broader institutional improvements, serving as a model for other universities and organizations seeking to implement effective HR management systems.

1.7 Limitations of the Project

This project is developed within defined academic, technical, and financial constraints. While it addresses major operational problems in human resource management at Bahir Dar University, it does not cover all HR functions.

The system solves the organization's core problems related to employee clearance, leave management, payroll name transfer, and internal job vacancy dissemination by automating these processes and reducing dependence on manual procedures. Within the given project timeframe, the main deliverables include system requirement analysis, design documentation, implementation of core HR modules, and system testing.

However, several challenges limited the full achievement of the initial plan. These include limited development time, zero-budget constraints, and restricted technical capacity. As a result, some advanced HR functionalities were excluded.

The project does not include an AI-based chatbot due to the unavailability of reliable free AI APIs, as premium services require additional costs. Mobile application development is also excluded because the system is intended to operate within the university intranet, where employees access services using desktop computers. Biometric attendance management is not implemented due to the complexity of integration, hardware costs, potential knowledge gaps, and time limitations. Furthermore, advanced analytics, reporting tools, and employee performance review modules are not developed due to insufficient time and expertise.

Activities left undone due to these constraints include:

- Integration of AI-powered chatbot services
- Development of a mobile-based HR application
- Implementation of biometric attendance systems
- Development of advanced analytics, reporting, and performance evaluation modules
- Implementation of full recruitment management features beyond job posting

1.8 Scope of the Project

The scope of this project defines the functional boundaries of the developed Human Resource Management System. The system is limited to the implementation of selected HR business processes that address the most critical challenges faced by Bahir Dar University.

The HRMS includes employee clearance management, electronic leave management, digital employee name transfer for payroll processing, and internal job posting and vacancy announcement. It also provides centralized employee data management and role-based access control to ensure system security and proper authorization.

The system does not extend to attendance tracking, performance management, training management, or external recruitment platforms. All system operations are designed to function within the university intranet environment.

1.9 Organization of the Project

This project is organized into several chapters, each addressing a specific aspect of the study and system development.

Chapter One

This chapter introduces the Human Resource Management System (HRMS) project for Bahir Dar University. It outlines the project's background, the problem of inefficient manual HR processes, and the core objective to automate clearance, leave, and payroll workflows. The methodology used for requirement gathering and system design is summarized. It also covers the project's feasibility, its significance to various stakeholders, and the defined scope and limitations, such as the exclusion of mobile apps and AI features due to time and budget constraints. **Chapter Two** reviews related literature, including concepts of human resource management systems, clearance and leave management processes, and relevant technologies.

Chapter Two

This chapter details the system's requirements and design. It contrasts the inefficient existing HR system with the proposed centralized web-based HRMS. It covers functional requirements (e.g., secure login, automated workflows) and non-functional needs (performance, security). Key sections include

use case models, business rules, and visual designs like UI prototypes and UML diagrams (activity, sequence, class models) for processes like registration and multi-campus clearance. System requirements (hardware/software) and future change cases for potential enhancements are also identified.

Chapter Three

This chapter details the system architecture. It covers the high-level component and deployment modeling, showing how the frontend (React), backend (Node.js), and database (PostgreSQL) are structured and deployed. The detailed design includes a refined class model and persistent database schema to support HR data. User interface designs from earlier prototypes are finalized for development. Access control and security mechanisms such as role-based authentication, encryption, and secure APIs are specified to ensure system integrity and data protection across all university campuses.

Chapter Four

This chapter describes the development phase using the chosen tech stack. Core modules such as authentication, employee self-registration, leave management, and digital clearance workflows are built and integrated. API endpoints connect the frontend and backend, while the PostgreSQL database stores all HR records. The implementation addresses real-time updates, multi-campus routing for approvals, and role-based dashboards. Challenges in data migration, integration, and ensuring system performance are discussed, along with the solutions applied to deliver a functional HRMS portal.

Chapter Five

This chapter covers the validation of the HRMS. Testing includes unit, integration, system, and user acceptance testing to verify all functionalities like login, leave requests, and clearance workflows. Performance and security tests ensure the system meets speed, scalability, and safety requirements. Feedback from HR staff and employees is gathered to assess usability and effectiveness. The evaluation confirms that the system meets its objectives, highlights observed limitations, and provides recommendations for future enhancements before deployment.

Chapter Two: System Features

2.1 The Existing System

The existing Human Resource Management system at Bahir Dar University operates predominantly through manual and semi-automated procedures. Although the university previously introduced a digital HR system, it is currently non-operational due to hardware failure. As a result, most human resource activities are conducted using paper-based documents and fragmented electronic files, such as standalone spreadsheets, across different campuses, departments, and administrative units. This operational environment leads to fragmented, slow, and inefficient HR service delivery.

In the current system, employee records such as personal data, employment history, leave information, and clearance documents are managed manually. When an employee is hired, their information is collected using paper forms and stored in physical files. Some data are captured in spreadsheets, but these files are decentralized, not updated in real time, and inaccessible across departments. Retrieving employee data requires searching through physical files or isolated electronic documents, making the process slow, labor-intensive, and error-prone.

Daily administrative processes such as leave management are also handled manually. Employees submit leave requests using physical forms, which require manual approval by supervisors before being forwarded to the HR office. HR officers calculate leave balances manually and record approvals in logs or spreadsheets. Employees have no visibility into their leave balances or request status, creating additional administrative workload and delays in service delivery.

Sabbatical leave management follows similar manual verification procedures. Academic staff meet eligibility criteria (e.g., one year of sabbatical after seven years of service) based on HR officers' manual verification of service years. This verification requires reviewing employment records and is susceptible to delays, data inconsistencies, and human error.

The employee clearance process is another critical function executed through physical paperwork. Departing employees must move from office to office to obtain clearance signatures from multiple units. Each unit verifies outstanding obligations independently and signs the clearance form. Once signatures are collected, the employee returns the document to the HR office for final approval and

archiving. There is no centralized tracking mechanism, which results in processing delays, repeated follow-ups, misplaced documents, and a lack of transparency in clearance status.

Payroll-related communication between the HR office and Finance department is semi-automated but still inefficient. HR compiles employee payroll information manually and transmits it using spreadsheets or paper documents. Errors or omissions are common and often lead to salary delays, repeated corrections, and dissatisfied employees who must follow up with both HR and Finance due to the absence of an integrated communication channel.

Beyond hardware failure, several key drawbacks characterize the existing system:

1. Manual, paper-based workflows: These create delays, increase administrative workload, and hinder timely service delivery.
2. Lack of system integration: HR processes operate in isolation, resulting in inconsistent data, redundant work, and weak cross-departmental coordination.
3. Absence of centralized and real-time information access: Personnel data are stored in scattered files, making verification and retrieval difficult and error-prone.
4. Limited transparency and tracking capability: Employees cannot independently track clearance, leave balances, or payroll status, leading to repeated inquiries and follow-ups.
5. High risk of data errors, loss, or duplication: Physical documents and isolated spreadsheets are vulnerable to misplacement, unauthorized modification, and inconsistencies.
6. Insufficient communication mechanisms between HR and Finance: Payroll processes rely on manual exchanges that introduce delays and operational risk.
7. Poor scalability and maintainability: Manual procedures are difficult to scale across multiple campuses and administrative units, particularly as personnel numbers increase.
8. Service quality limitations: HR service delivery suffers from delays, data insecurity, and reduced accountability, negatively affecting employee experience and institutional efficiency.

These limitations collectively highlight the need for a centralized, automated, and reliable Human Resource Management System capable of streamlining workflows, enhancing data integrity, and improving service quality across Bahir Dar University.

2.2 The Proposed System

The proposed system is a centralized, web-based Human Resource Management System designed to overcome the limitations of the existing manual and semi-automated HR processes at Bahir Dar University. The system aims to automate key HR functions, reduce paper-based operations, improve data accuracy, and enhance transparency and efficiency across all campuses and departments. The proposed HRMS focuses on employee clearance management, leave management, sabbatical leave processing, and digital employee name transfer for payroll purposes.

In the proposed system, all employee data is stored in a centralized and secure database that can be accessed in real time by authorized users. Role-based access control ensures that HR officers, supervisors, finance staff, and employees can only access information relevant to their responsibilities. This centralized data management approach eliminates fragmented records and improves coordination between departments.

The employee clearance process is fully automated through a digital workflow. When a clearance request is initiated, the system automatically routes the request to relevant departments for approval. Each department reviews and approves clearance requests electronically, and the system updates the clearance status in real time. Employees and HR officers can track clearance progress without physical document movement, reducing delays and preventing document loss.

The proposed system introduces an electronic leave management module that allows employees to submit leave requests online. Supervisors and HR officers review and approve requests through the system, and leave balances are automatically calculated and updated. Employees can view their leave history, approval status, and remaining leave days at any time, improving transparency and reducing unnecessary communication.

In addition, the proposed HRMS includes an automated sabbatical leave management feature. When an academic staff member submits a sabbatical leave request, the system automatically checks the employee's length of service using their employment start date stored in the database. If the employee

has completed seven years of service, the system allows the sabbatical request to proceed through the approval workflow. If the employee has not met the seven-year requirement, the system automatically rejects the request and notifies the employee with a clear explanation. This automation ensures consistent enforcement of university policy and eliminates manual eligibility verification.

To address payroll-related challenges, the proposed system includes a secure digital employee name transfer mechanism that allows HR officers to electronically transmit approved payroll information to the Finance department. This integration minimizes errors, reduces processing time, and ensures timely salary payments.

Overall, the proposed Human Resource Management System provides a secure, efficient, and transparent platform for managing HR operations at Bahir Dar University. By automating clearance, leave, sabbatical, and payroll-related processes, the system reduces administrative workload, enhances accountability, improves data security, and modernizes human resource management practices in line with institutional goals.

2.3 Requirement Analysis

2.3.1 Functional Requirements

Table 2.1:Functional Requirements

Requirement ID	Requirement Name	Description	Priority
FREQ-1	Secure login and user authentication	The system will require secure login functionality to authenticate users, ensuring that only authorized personnel (HR staff, administrators, or managers) can access employee records.	High
FREQ-2	Role-based access control for employee data	Role-based access control (RBAC) will ensure that users can only access employee data relevant to their role, maintaining confidentiality and privacy.	High
FREQ-3	Centralized employee	Employee records (personal details,	High

	record management	employment history, documents) will be centralized in a unified database, ensuring consistency and easy access for HR staff and managers across campuses.	
FREQ-4	Real-time data synchronization	Employee data, leave balances, and permissions will be synchronized in real-time across all departments and campuses, ensuring accurate and up-to-date information.	High
FREQ-5	Integration with national ID and university ID systems	Integration with national and university ID systems will ensure accurate identification and streamlined data entry for employees.	Medium
FREQ-6	Secure data sharing between departments and campuses	The system will allow secure sharing of employee records and approvals across multiple campuses with proper access control.	High
FREQ-7	Web-based access for employees and HR staff	Employees and HR staff can access the system through a secure web interface for tasks like leave requests, birth permission, annual permission, and clearance approvals.	High
FREQ-8	Generate real-time HR reports and analytics	The system will generate real-time reports on employee leave, permissions, clearances, and other HR metrics to support informed decision-making by HR and administrators.	Medium
FREQ-9	Automated workflow for clearance, leave, and permissions	The system will provide automated workflows for clearance sign-offs, birth permissions, and annual leave approvals, removing the need for physical paperwork and multi-campus routing.	High

FREQ-10	System scalability for handling increasing HR data	The system will be designed to scale efficiently, handling increasing numbers of employees, documents, and administrative requests without compromising performance.	High
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2.3.2 System Use case

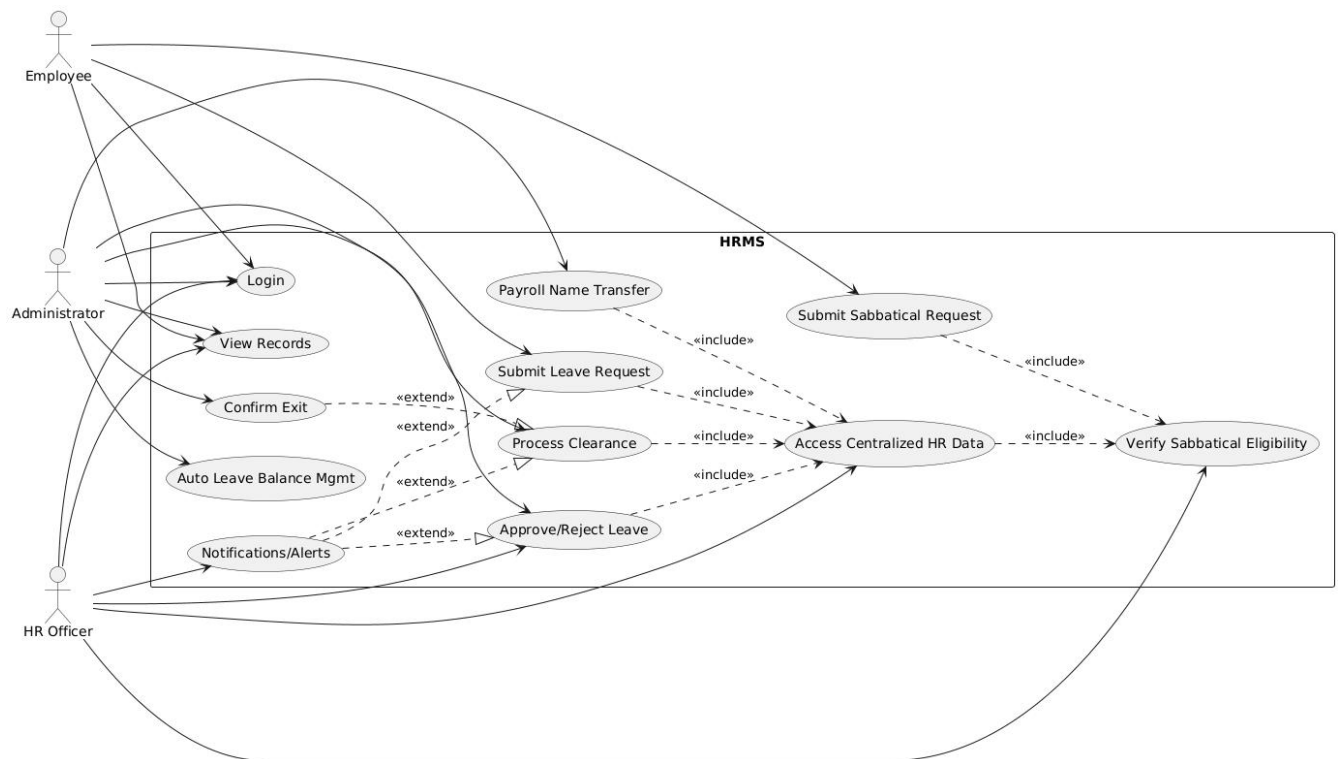


Figure 2.1: Use case diagram

2.3.2.1 Use Case documentation

Table UC-001: Login Use Case

Field	Description
Use Case Number	UC-001
Use Case Name	Login
Actor(s)	Employee, HR Officer, Administrator

Field	Description
Description	Allows actors to authenticate into the system and gain access based on their assigned roles.
Pre-Condition	Actor has registered credentials stored in system.
Post-Condition	Actor is authenticated and redirected to respective dashboard.
Priority	High
Basic Course of Action	1. Actor navigates to login page. 2. Actor enters username and password. 3. System validates credentials. 4. System grants access.
Alternate Course of Action	A1. Invalid credentials: System displays error. A2. Account locked: Actor instructed to contact HR.

Table UC-002: Submit Clearance Request Use Case

Field	Description
Use Case Number	UC-002
Use Case Name	Submit Clearance Request
Actor(s)	Employee
Description	Allows employees to submit clearance requests (resignation, transfer, contract completion, etc.) electronically.
Pre-Condition	Employee is authenticated and clearance type is applicable.

Field	Description
Post-Condition	Clearance request stored and routed for approval.
Priority	High
Basic Course of Action	1. Employee opens clearance module. 2. Employee selects clearance type. 3. Employee enters details. 4. System validates input. 5. System submits request and forwards for processing.
Alternate Course of Action	A1. Missing/invalid data: System requests correction. A2. Ineligible clearance type: System rejects request with explanation.

Table UC-003: Process Clearance Request Use Case

Field	Description
Use Case Number	UC-003
Use Case Name	Process Clearance
Actor(s)	HR Officer, Administrator
Description	Allows authorized actors to approve, reject, or comment on clearance requests.
Pre-Condition	Clearance request exists in pending state and actor has approval rights.
Post-Condition	Clearance status updated and stored in centralized database.
Priority	High

Field	Description
Basic Course of Action	1. Actor views pending requests. 2. Actor selects request. 3. Actor reviews employee information. 4. Actor approves, rejects, or comments. 5. System updates status and notifies employee.
Alternate Course of Action	A1. Rejection: System notifies employee with justification. A2. Escalation: System forwards to Administrator.

Table UC-004: Submit Leave Request Use Case

Field	Description
Use Case Number	UC-004
Use Case Name	Submit Leave Request
Actor(s)	Employee
Description	Allows employees to submit digital leave requests.
Pre-Condition	Employee authenticated and leave type applicable.
Post-Condition	Leave request stored and forwarded for approval.
Priority	High
Basic Course of Action	1. Employee opens leave module. 2. Employee enters leave type and dates. 3. System validates data. 4. Employee submits request. 5. System routes request for approval.

Field	Description
Alternate Course of Action	A1. Invalid date range: System prompts correction. A2. Leave type not permitted: System blocks request.

Table UC-005: Approve/Reject Leave Request Use Case

Field	Description
Use Case Number	UC-005
Use Case Name	Approve/Reject Leave Request
Actor(s)	HR Officer, Administrator
Description	Allows authorized actors to approve, reject, or modify leave requests.
Pre-Condition	Leave request exists and is pending approval.
Post-Condition	Status updated, employee notified, leave balance recalculated.
Priority	High
Basic Course of Action	1. Actor views pending leave requests. 2. Actor reviews leave details. 3. Actor selects approve/reject/modify. 4. System updates status and notifies employee.
Alternate Course of Action	A1. Rejection: System informs employee with reason. A2. Modification: System requests employee acknowledgment.

Table UC-006: Verify Sabbatical Eligibility Use Case

Field	Description
Use Case Number	UC-006
Use Case Name	Verify Sabbatical Eligibility
Actor(s)	HR Officer, Administrator
Description	Automatically verifies whether an academic employee meets sabbatical requirements.
Pre-Condition	Sabbatical request submitted and employment data available.
Post-Condition	Eligible requests forwarded; ineligible requests rejected.
Priority	Medium
Basic Course of Action	1. System retrieves years of service. 2. System compares to sabbatical policy (e.g., 7 years). 3. System marks request as eligible. 4. System forwards for approval workflow.
Alternate Course of Action	A1. Eligibility not met: System rejects request and notifies employee.

Table UC-007: Payroll Name Transfer Use Case

Field	Description
Use Case Number	UC-007
Use Case Name	Payroll Name Transfer

Field	Description
Actor(s)	HR Officer, Administrator
Description	Enables secure transfer of employee payroll information for salary processing.
Pre-Condition	Payroll data validated and approval rights granted.
Post-Condition	Payroll data transmitted successfully for payment processing.
Priority	High
Basic Course of Action	1. HR Officer prepares payroll data. 2. System validates data. 3. System transmits data securely. 4. Administrator confirms receipt.
Alternate Course of Action	A1. Validation failure: System requests correction. A2. Transmission failure: System logs and alerts actor.

Table UC-008: View HR Records Use Case

Field	Description
Use Case Number	UC-008
Use Case Name	View HR Records
Actor(s)	Employee, HR Officer, Administrator
Description	Enables viewing of clearance, leave, sabbatical, and payroll records.

Field	Description
Pre-Condition	Actor authenticated and has view permissions.
Post-Condition	Records displayed without modification.
Priority	Medium
Basic Course of Action	1. Actor selects record type. 2. System retrieves relevant data. 3. System displays data.
Alternate Course of Action	A1. No data: System displays empty result message.

Table UC-009: Notifications & Alerts Use Case

Field	Description
Use Case Number	UC-009
Use Case Name	Notifications & Alerts
Actor(s)	Employee, HR Officer, Administrator
Description	Sends alerts about approvals, rejections, workflow tasks, and updates.
Pre-Condition	Triggering event executed in workflow.
Post-Condition	Notification delivered and logged.
Priority	Medium
Basic Course of Action	1. System detects event. 2. System generates notification. 3. Actor receives notification.

Field	Description
Alternate Course of Action	A1. Actor offline: Notification queued for next login.

2.3.3 Business Rule Documentation

Table 2.2: Business Rule Documentation

Identifier	Name	Description	Reference	Revision History
BR-01	Employee Registration Control	Only HR officers are allowed to create, update, or deactivate employee records in the system.	BDU HR Policy	Version 1.0
BR-02	Role-Based Access	Users can access system functions only according to their assigned roles.	BDU ICT & HR Security Guidelines	Version 1.0
BR-03	Leave Approval Flow	Leave requests must be approved by the immediate supervisor before HR approval.	BDU Leave Administration Rules	Version 1.0
BR-04	Leave Balance Update	Leave balances are automatically updated after leave approval or rejection.	BDU Leave Management Procedure	Version 1.0
BR-05	Sabbatical Eligibility Rule	Sabbatical leave is allowed only after completing seven years of service.	BDU Academic Staff Regulation	Version 1.0
BR-06	Sabbatical Request Validation	Sabbatical requests are rejected if the seven-year service requirement is not met.	BDU Sabbatical Leave Policy	Version 1.0

Identifier	Name	Description	Reference	Revision History
BR-07	Clearance Completion Rule	Clearance is valid only when all required departmental approvals are completed.	BDU Employee Clearance Guideline	Version 1.0
BR-08	Payroll Clearance Dependency	Employees with incomplete clearance are excluded from payroll name transfer.	BDU HR–Finance Policy	Version 1.0
BR-09	Payroll Name Transfer Authority	Only HR officers can approve and send payroll name lists to Finance.	BDU Payroll Processing Rules	Version 1.0

2.3.4 User Interface prototype

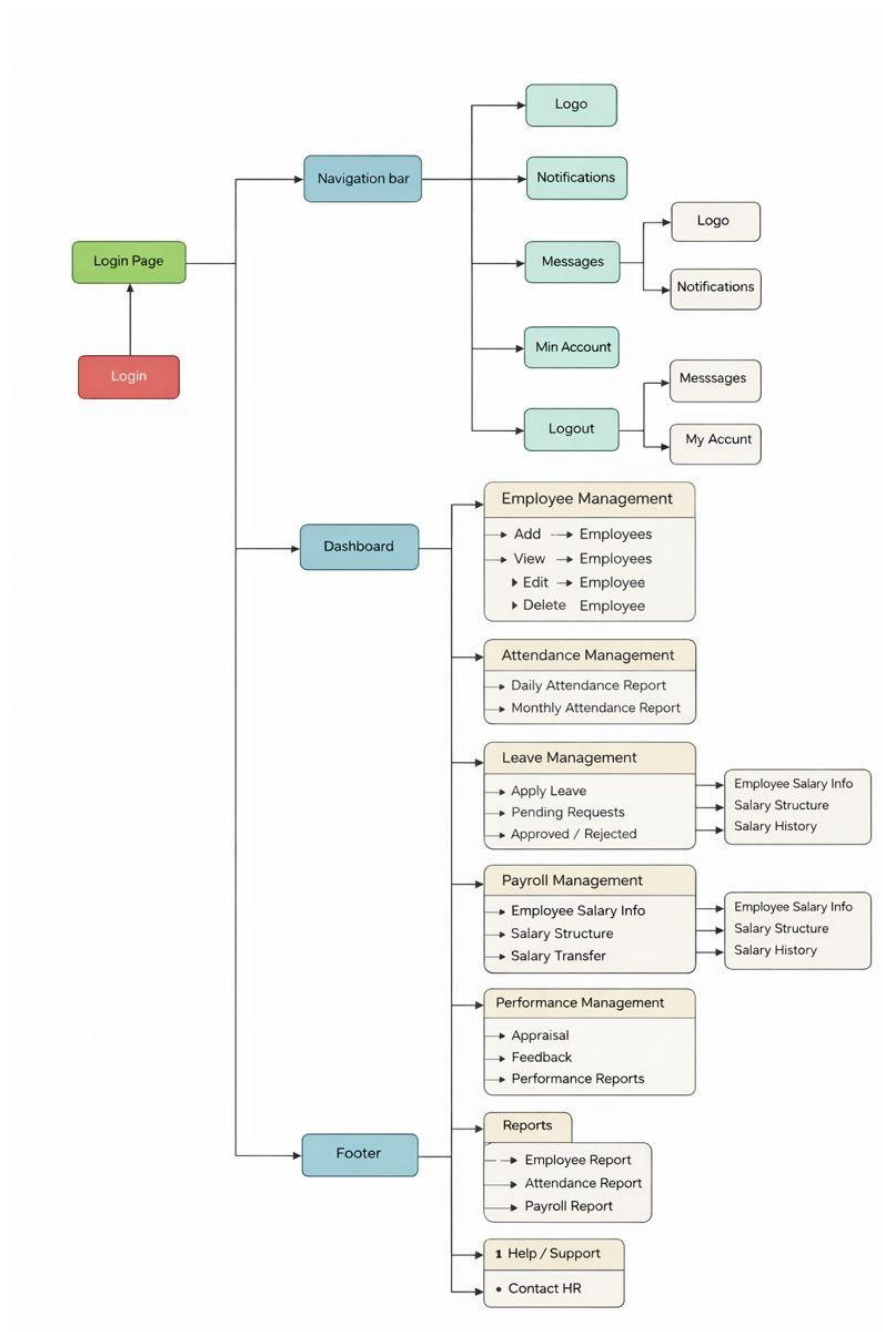


Figure 2.2: Home Screen

HRMS - EMPLOYEE SIDE

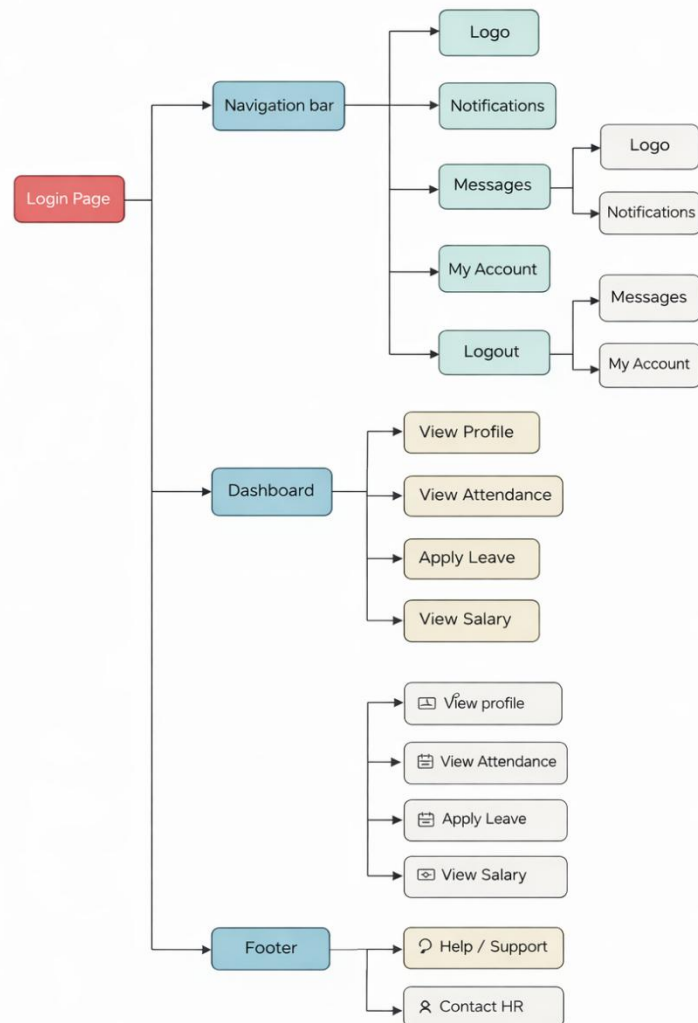


Figure 2.3:Employee Dashboard

2.3.5 State chart diagram

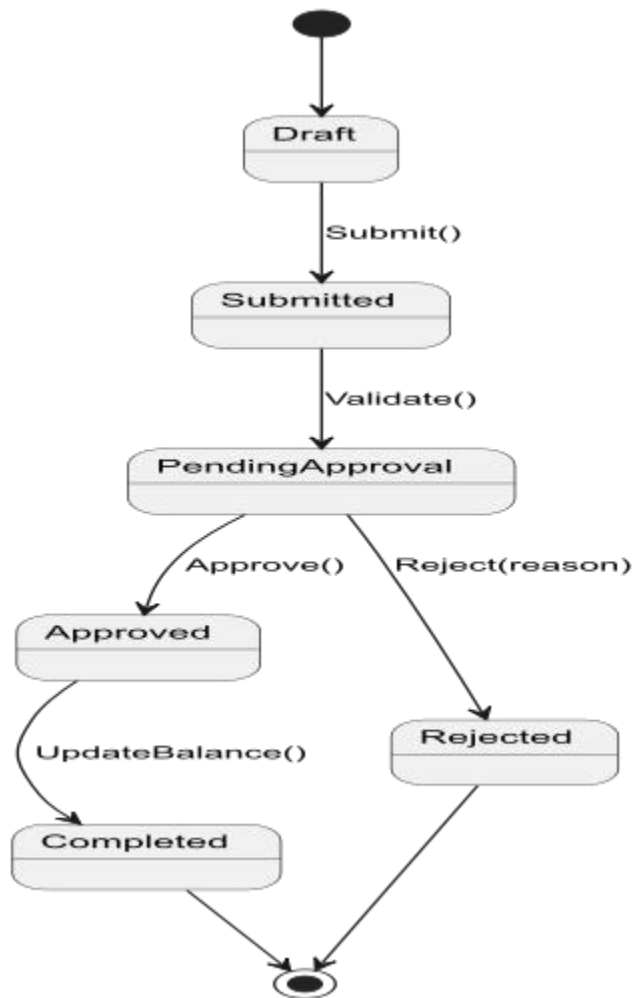


Figure 2.4: State Chart: Leave Request

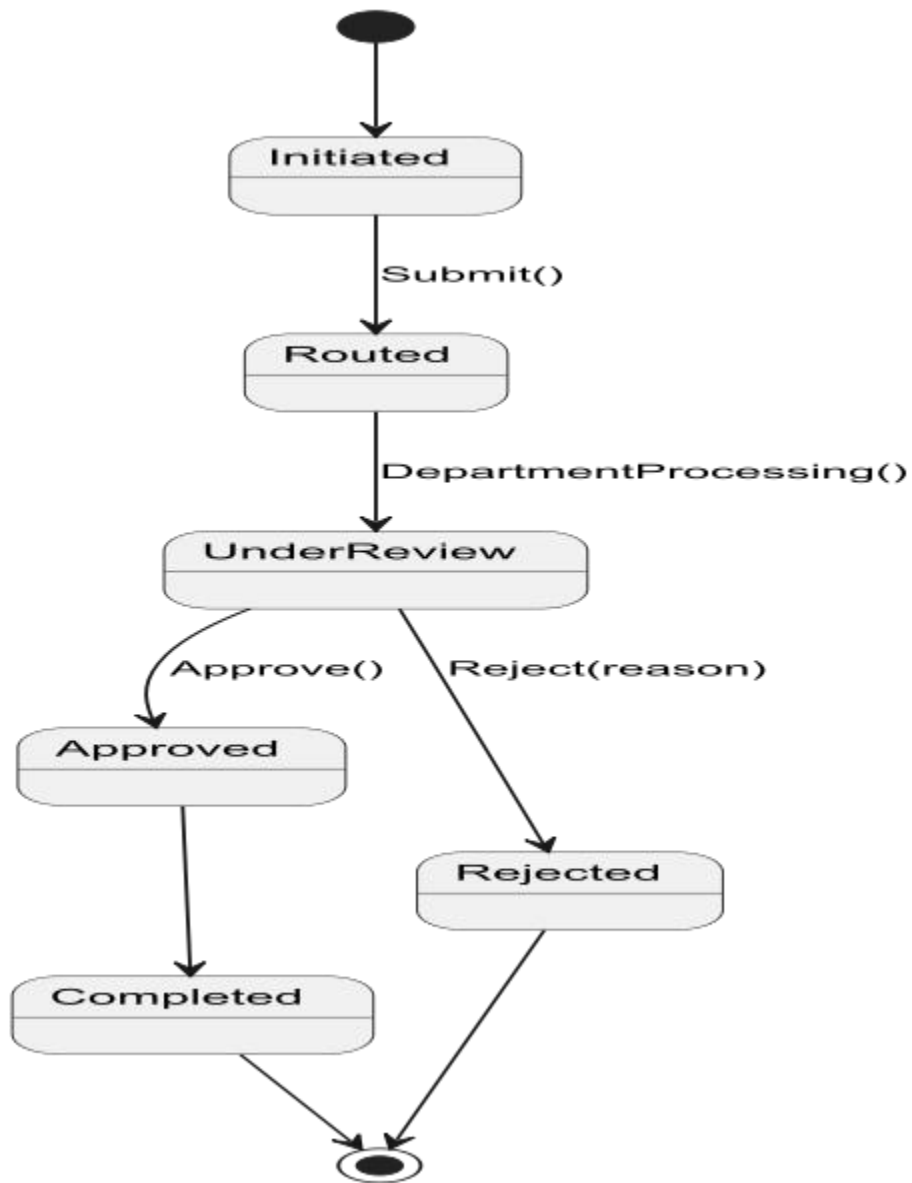


Figure 2.5: Clearance Request state chart diagram

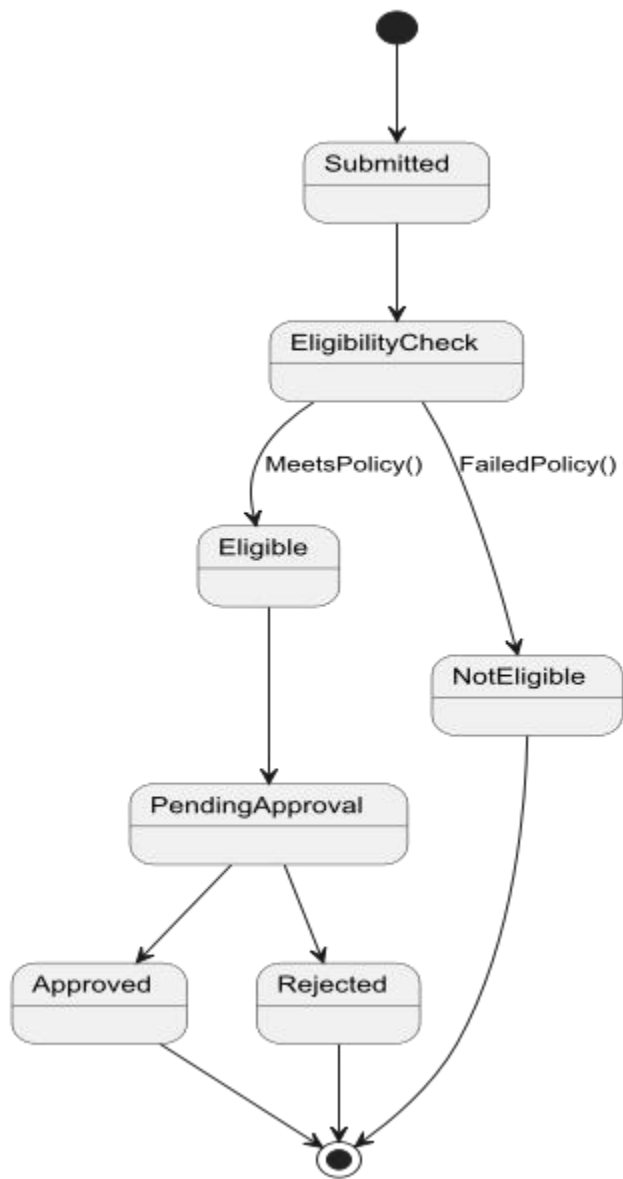


Figure 2.6: Sabbatical Request state chart diagram

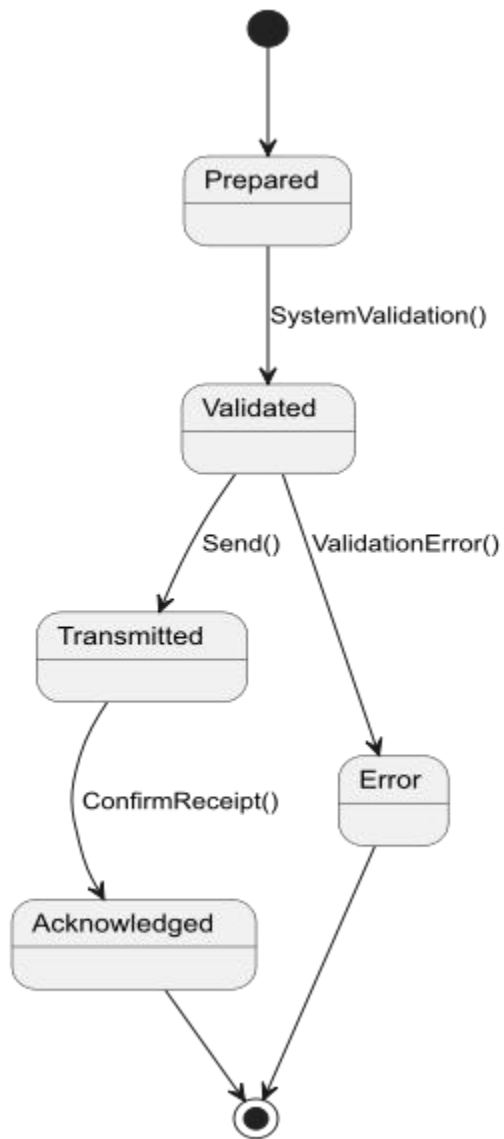


Figure 2.7: Payroll Transfer

2.3.6 Activity Diagram

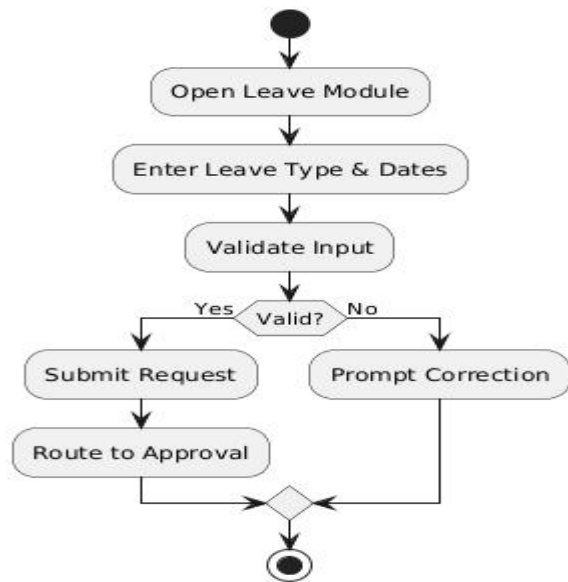


Figure 2.8: Submit Leave Request

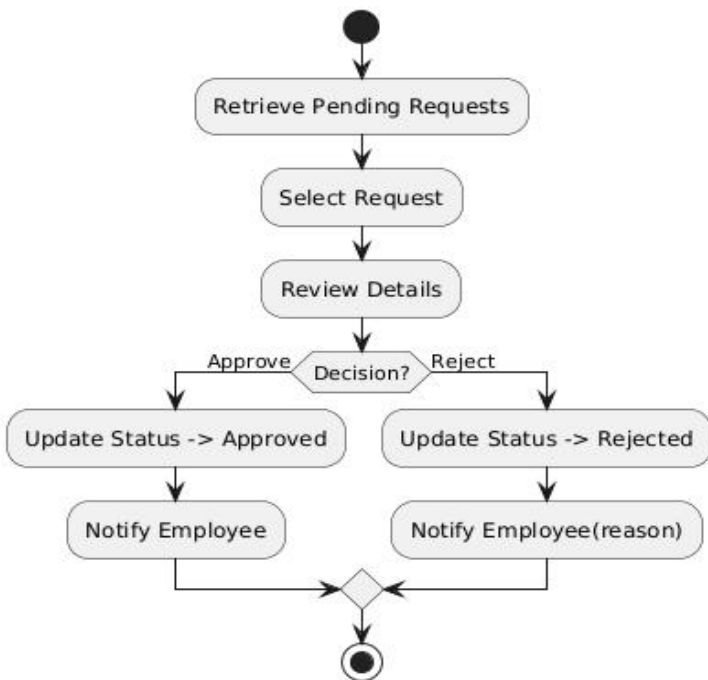


Figure 2.9: Process Clearance

2.3.7 Sequence diagram

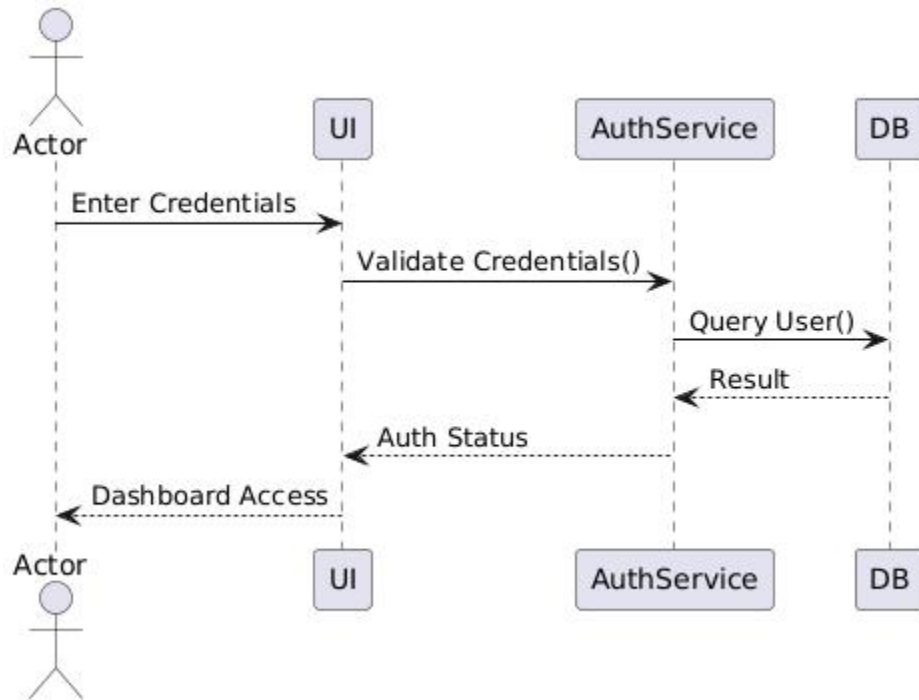


Figure 2.10: Login Sequence

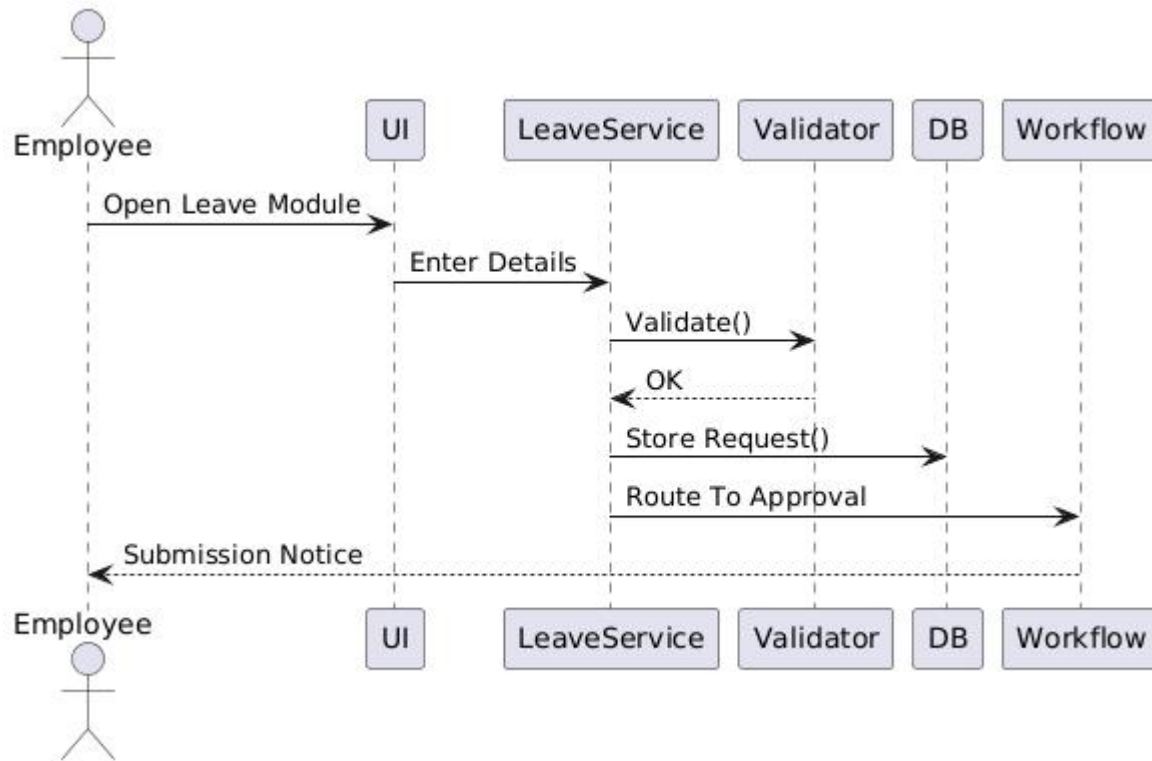


Figure 2.11: Submit Leave Request Sequence

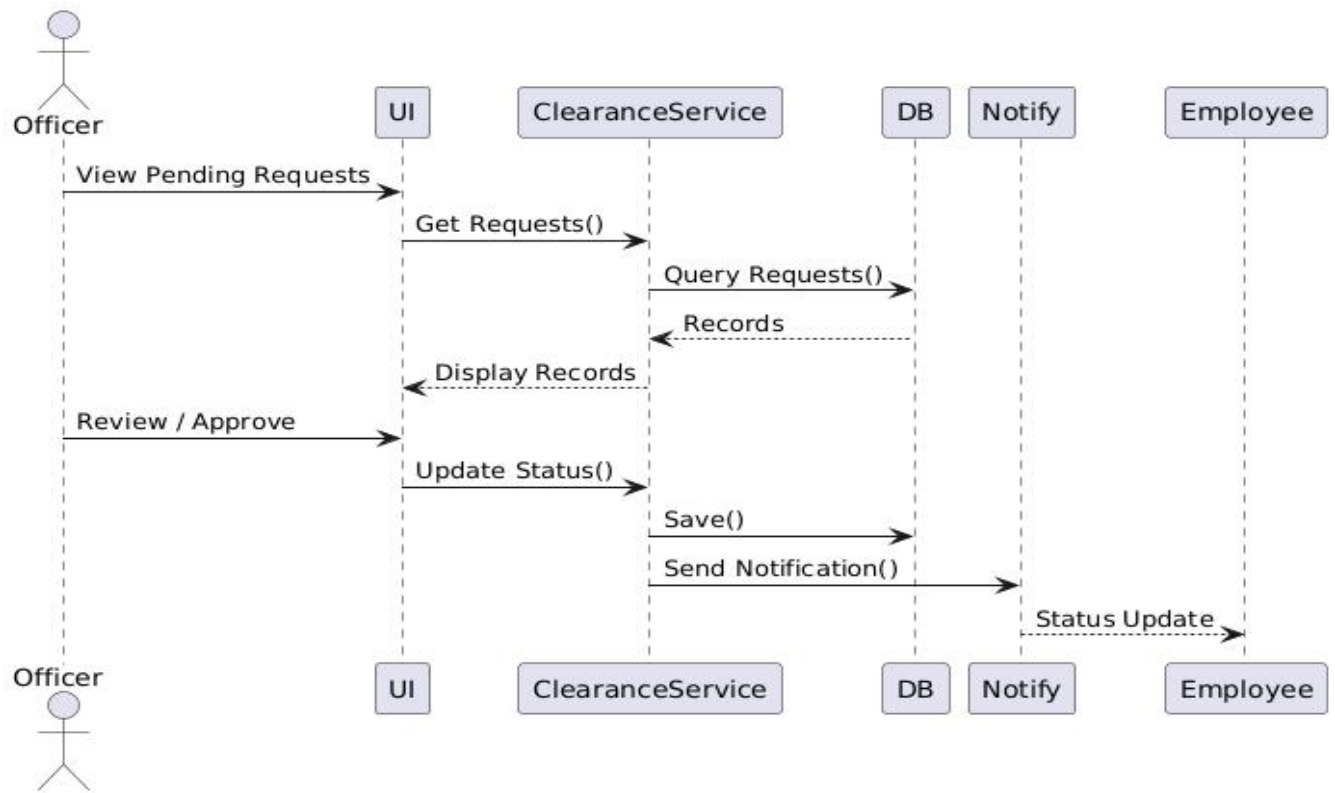


Figure 2.12: Process Clearance Request Sequence

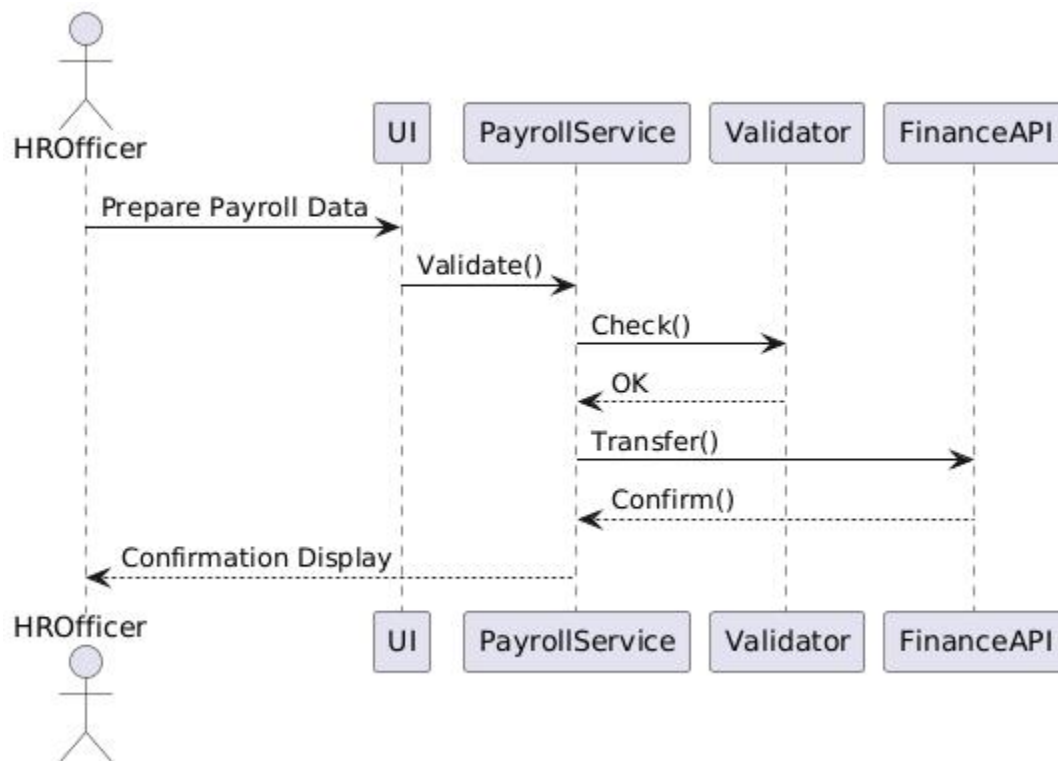


Figure 2.13: Payroll Name Transfer Sequence

2.3.8 Analysis Class Model

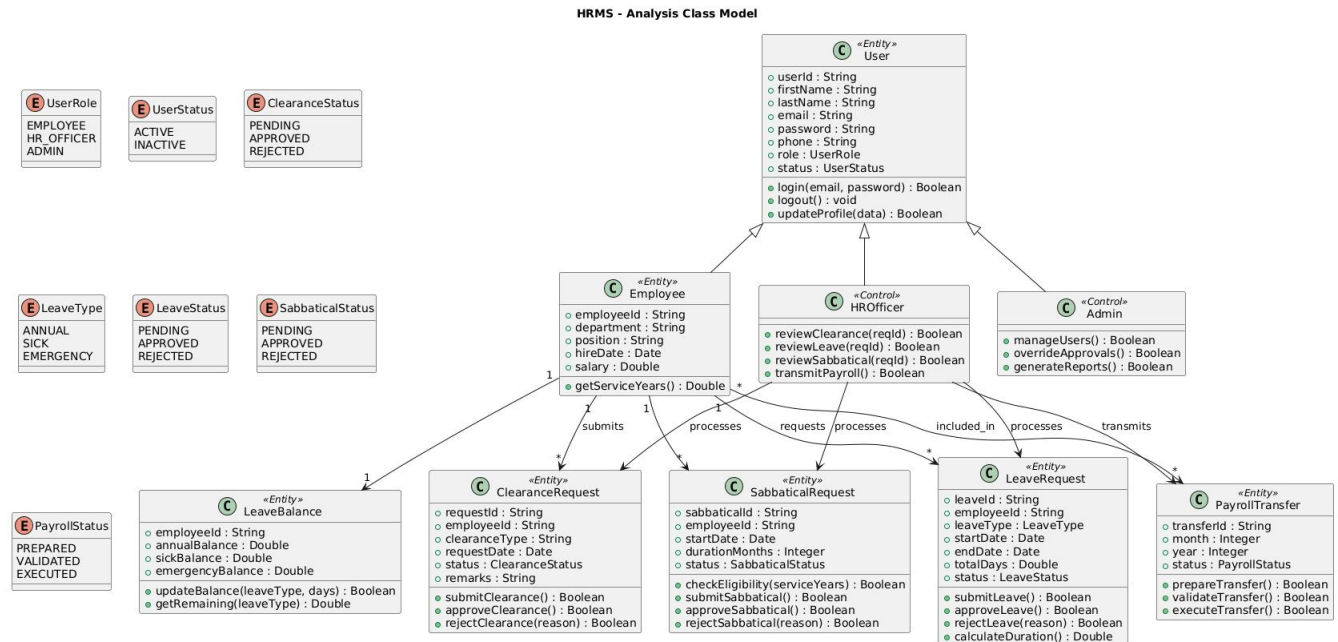


Figure 2.14: Analysis Class Model

2.3.9 Logic Model

1.Login Process

Algorithm Login(email, password)

BEGIN

 READ email, password

 FETCH userRecord WHERE email = input.email

 IF userRecord = NULL THEN

 DISPLAY "User Not Found"

 RETURN LoginFailed

 ENDIF

 IF userRecord.password != password THEN

 DISPLAY "Invalid Credentials"

 RETURN LoginFailed

```

ENDIF

IF userRecord.status = INACTIVE THEN

    DISPLAY "Account Disabled"

    RETURN LoginFailed

ENDIF

SET session.user = userRecord

DISPLAY "Login Successful"

RETURN LoginSuccess

END

```

2. Clearance Request Process

```

Algorithm SubmitClearanceRequest(employeeId, clearanceType)

BEGIN

    VALIDATE clearanceType

    CREATE ClearanceRequest WITH status = PENDING

    FOR each department in RequiredDepartments DO

        SEND ClearanceApprovalRequest TO department

    END FOR

    DISPLAY "Clearance Request Submitted"

END

```

Clearance Approval Workflow

```

Algorithm ProcessClearanceApproval(requestId, departmentDecision)

BEGIN

    FETCH ClearanceRequest(requestId)

```

```

IF departmentDecision = REJECTED THEN

    UPDATE ClearanceRequest.status = REJECTED

    SEND Notification TO employee

    EXIT

END IF

MARK DepartmentApprovalStatus = APPROVED

IF all departments APPROVED THEN

    UPDATE ClearanceRequest.status = APPROVED

    SEND Notification TO employee

END IF

END

```

3. Leave Request Process

Algorithm SubmitLeaveRequest(employeeId, leaveType, startDate, endDate)

```

BEGIN

    totalDays = CalculateDays(startDate, endDate)

    FETCH LeaveBalance(employeeId)

    IF LeaveBalance(leaveType) < totalDays THEN

        DISPLAY "Insufficient Leave Balance"

        RETURN RequestFailed

    ENDIF

    CREATE LeaveRequest WITH status = PENDING

    SEND Notification TO HROfficer

    DISPLAY "Leave Request Submitted"

```

END

4. Leave Approval + Balance Update

Algorithm ApproveLeaveRequest(leaveId)

BEGIN

 FETCH LeaveRequest(leaveId)

 FETCH LeaveBalance(employeeId)

 UPDATE LeaveRequest.status = APPROVED

 UPDATE LeaveBalance(leaveType) = LeaveBalance - totalDays

 SEND Notification TO employee

END

Alternate (Rejection)

Algorithm RejectLeaveRequest(leaveId, reason)

BEGIN

 FETCH LeaveRequest(leaveId)

 UPDATE LeaveRequest.status = REJECTED

 SEND Notification TO employee WITH reason

END

5. Sabbatical Leave (Eligibility Rule)

Algorithm SubmitSabbaticalRequest(employeeId, startDate, duration)

BEGIN

 FETCH Employee(employeeId)

 years = CalculateServiceYears(Employee.hireDate)

```

IF years < 7 THEN

    DISPLAY "Not Eligible for Sabbatical"

    RETURN RequestFailed

ENDIF

CREATE SabbaticalRequest.status = PENDING

ROUTE request TO HROfficer and Admin

SEND Notification

END

```

Approval Workflow

```

Algorithm ProcessSabbaticalApproval(requestId, decision)

BEGIN

    IF decision = REJECTED THEN

        UPDATE SabbaticalRequest.status = REJECTED

        SEND Notification

        EXIT

    ENDIF

    IF decision = APPROVED AND all approvals completed THEN

        UPDATE SabbaticalRequest.status = APPROVED

        SEND Notification

    ENDIF

END

```

6. Payroll Name Transfer Process

```

Algorithm ExecutePayrollTransfer(month, year)

```

```

BEGIN

    FETCH ApprovedEmployeeData(month, year)

    IF no data THEN

        DISPLAY "No Records Found"

        RETURN TransferFailed

    ENDIF

    VALIDATE DataIntegrity(ApprovedEmployeeData)

    IF validationFailed THEN

        DISPLAY "Validation Error"

        RETURN TransferFailed

    ENDIF

    SEND ApprovedEmployeeData TO Finance

    MARK TransferStatus = EXECUTED

    SEND Notification TO HR + Finance

END

```

7. Notification System

```

Algorithm SendNotification(userId, message)

BEGIN

    FETCH userContactInfo(userId)

    IF userContactInfo = NULL THEN

        EXIT

    ENDIF

    PUSH message TO NotificationQueue

```

END

2.4 Non-Functional Requirements

Our portal is designed to deliver reliable and efficient services to all users, including administrators, HR officers, and employees. In terms of performance, our portal should respond quickly to user actions such as logging in, viewing employee profiles, submitting leave requests, managing attendance, and generating reports. Under normal operating conditions, most operations should be completed within a few seconds. Our portal must be able to handle multiple users accessing the platform simultaneously without significant delays, and it should remain stable as the organization and number of records grow.

Reliability is a core requirement of our portal. The portal should be available at all times except during scheduled maintenance periods. It must process operations such as payroll handling, employee data updates, and attendance tracking accurately and consistently. Our portal should be able to recover from unexpected failures such as server crashes or power interruptions with minimal data loss. Regular automated backups should be performed to ensure that important information can be restored when necessary.

Security is a critical non-functional requirement since our portal manages sensitive personal and organizational data. All users must be authenticated before gaining access, and our portal must enforce role-based access control so that each user can only access features and information relevant to their responsibilities. Data transmitted between users and our portal should be encrypted, and sensitive information stored in the database must be protected against unauthorized access. Our portal should also include logging and monitoring mechanisms to detect suspicious activities and support auditing.

Usability is essential to ensure that our portal can be easily adopted by users with different technical backgrounds. The interface of our portal should be simple, intuitive, and consistent across all modules. Users should be able to complete common tasks without confusion or extensive training. Our portal should provide clear feedback, meaningful error messages, and guidance throughout important processes. It should also be responsive and accessible across different devices and screen sizes.

Maintainability is required so that our portal can evolve over time. The portal should be developed using modular and well-structured code to allow future enhancements, bug fixes, and system upgrades without major disruption. Proper documentation should be maintained so that developers can easily understand, maintain, and extend our portal. Finally, our portal should be compatible with modern web browsers and adaptable to different deployment environments. It should support integration with

external services such as email systems, reporting tools, or biometric and financial platforms if needed. This ensures that our portal remains flexible, scalable, and suitable for long-term organizational use.

2.5 System Requirements

This section presents the hardware and software requirements necessary for the successful deployment and operation of our portal.

2.5.1 Hardware Requirements

Our portal is a web-based system that operates in a client–server environment. For the server side, a machine with at least an Intel Core i5 processor or equivalent is required to efficiently process multiple user requests. A minimum of 8 GB RAM is recommended to support concurrent access, background services, and database operations. Storage of at least 256 GB SSD is required to store the operating system, application files, PostgreSQL database files, backups, and system logs. A reliable and stable internet connection is essential to ensure continuous availability of our portal. Cloud-based hosting environments may be used to improve scalability and availability.

On the client side, any modern computer, tablet, or smartphone capable of running a web browser can access our portal. A minimum of 4 GB RAM, a dual-core processor, and a stable internet connection are recommended. Optional devices such as printers and biometric attendance machines may be integrated depending on organizational needs.

2.5.2 Software Requirements

Our portal requires a modern operating system such as Windows 10 or later, macOS, or a recent Linux distribution on both server and client machines. On the server side, a backend runtime environment such as Node.js (version 18 or later) is required to run the application logic. PostgreSQL (version 13 or later) is used as the primary database management system for securely storing and managing employee records, attendance data, payroll information, and system logs.

Client-side access to our portal requires modern web browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari (latest versions), with support for HTML5, CSS3, and JavaScript. Additional development and maintenance tools include Visual Studio Code or a similar IDE, Git for version control, and Docker (optional) for containerized deployment. Our portal does not depend on special browser plug-ins, as all functionality is provided through standard web technologies.

2.6 Key abstraction with CRC analysis

2.6.1 Conceptual modeling : Class diagram

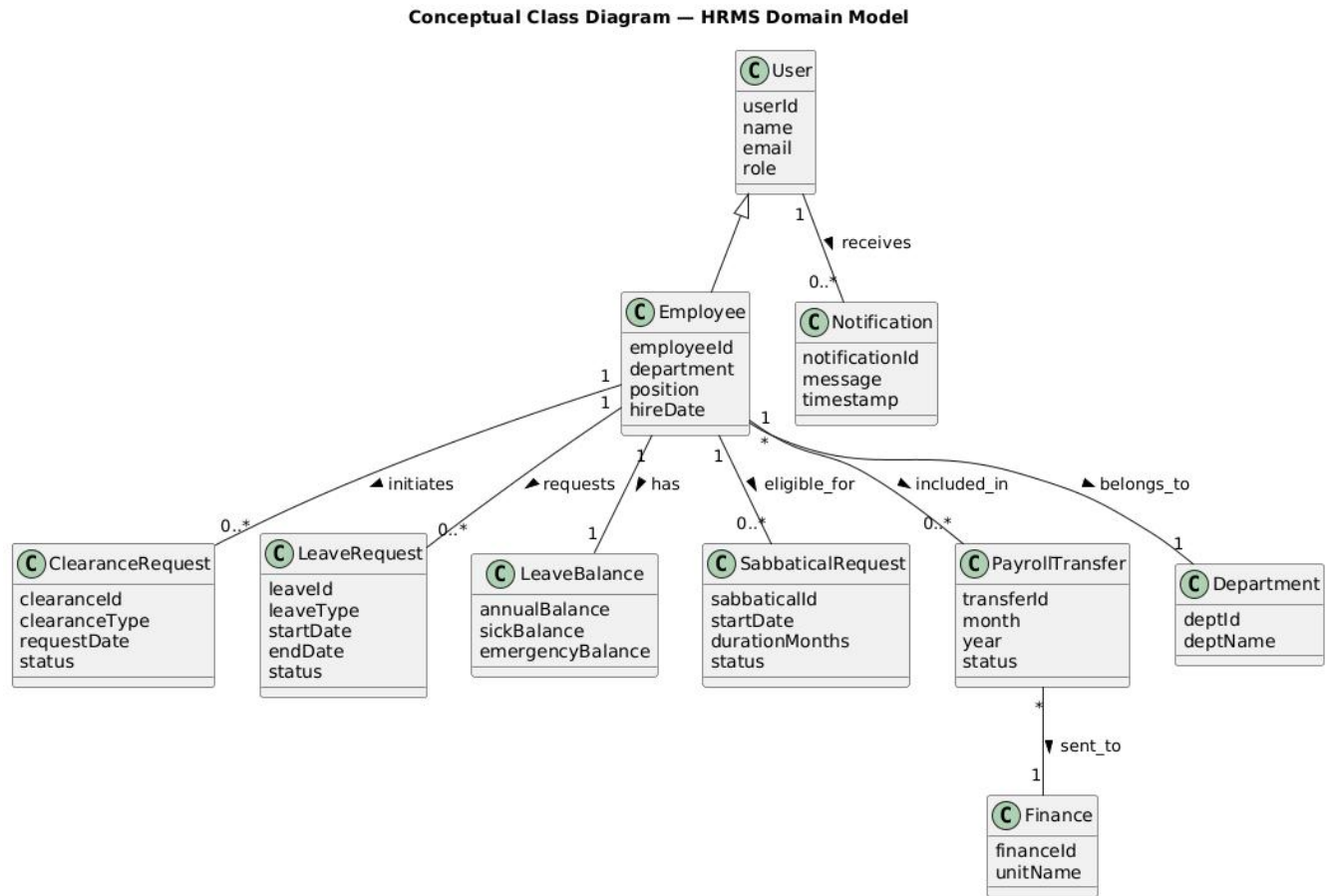


Figure2.15 :conceptual class diagram

2.6.2 Identifying Change Cases

Change Case: Instant Job Posting to Multiple Platforms

Change: Adding functionality that allows HR officers to instantly publish job vacancies to multiple recruitment platforms (e.g., university career portal, LinkedIn, Telegram, and internal bulletin boards) through a single interface.

Likelihood: Medium. Universities increasingly digitize recruitment workflows and external visibility of job postings is becoming essential.

Impact:

- High: Improves recruitment efficiency and widens applicant pool. Reduces repetitive manual posting work.
- Moderate: Requires API/configurable integration with third-party platforms and content formatting consistency.
- Low: Minimal disruption to internal HR workflows, but may require legal/administrative approvals for external dissemination.

Change Case: AI-Based Monthly and Annual HR Reporting

Change: Integrating AI-generated analytics dashboards capable of producing monthly and annual summaries for payroll, leave, clearance, employee turnover, and headcount data.

Likelihood: High. AI analytics adoption for administrative operations is accelerating, especially for data-driven decision-making.

Impact:

- High: Enhances strategic planning, improves decision-making, and reduces reporting workload for HR officers.
- Moderate: Requires integration of data pipelines, visualization engine, and model management.
- Low: Minimal direct interaction changes for employees; mostly affects HR management and administrative stakeholders.

Change Case: Integration with Finance System

Change: Transitioning from manual payroll export to automated or API-based integration between HRMS and Finance system for salary processing, allowances, deductions, and contract-based payments.

Likelihood: High. Financial automation and payroll integration are common modernization priorities in large institutions.

Impact:

- High: Eliminates manual data transfer errors, accelerates payroll cycles, and supports consistency between departments.
- Moderate: Requires architectural adjustments for secure communication, interface agreements, and audit trails.
- Low: Low impact on non-payroll HR processes but may require authentication enhancements and compliance checks

2.6.3 User Interface Prototyping

U

University HRMS

Human Resource Management System

Employee ID

Password

[Forgot password?](#)

Login

For support, contact HR at hr@university.edu

Figure 2.16 :login prototype UI

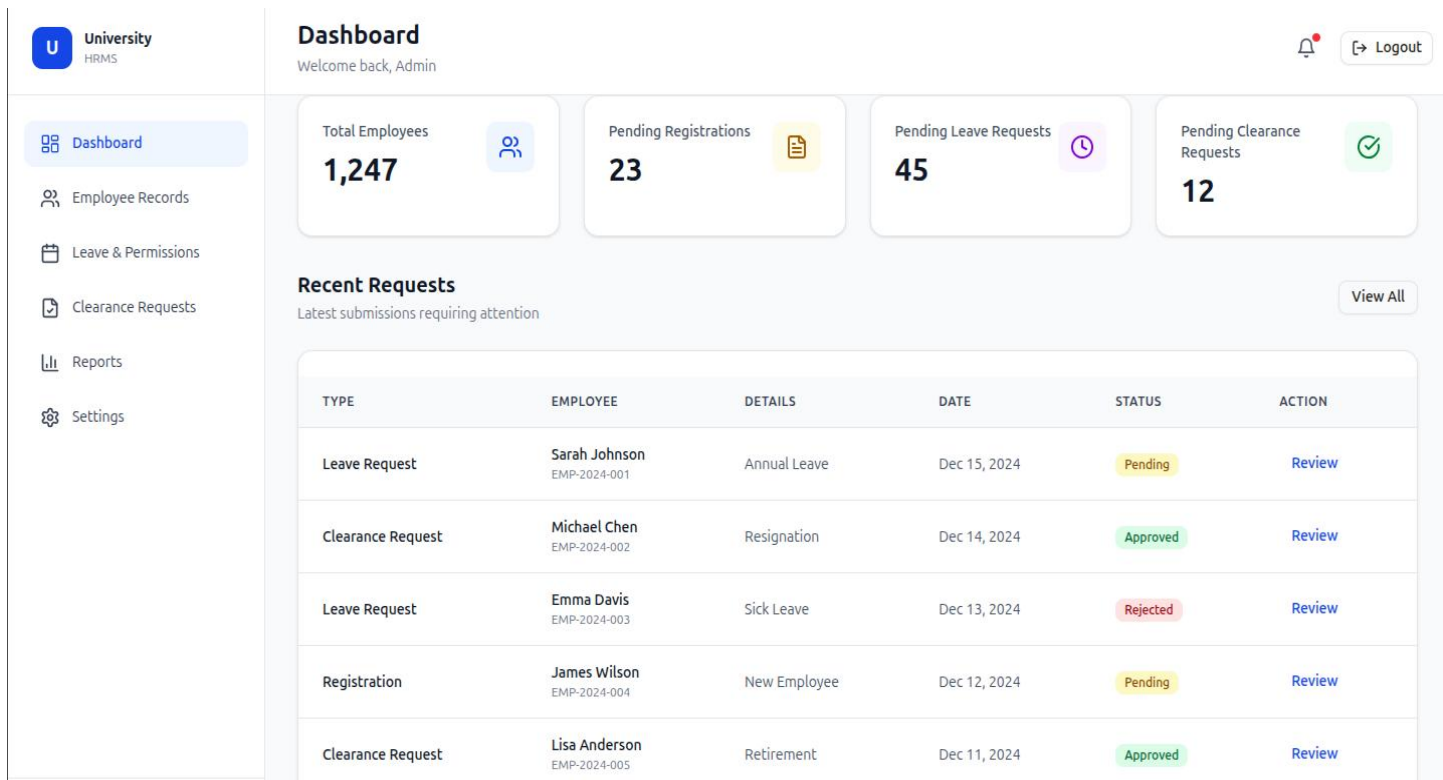


Figure 2.17 :Hr Admin Dashboard UI prototype

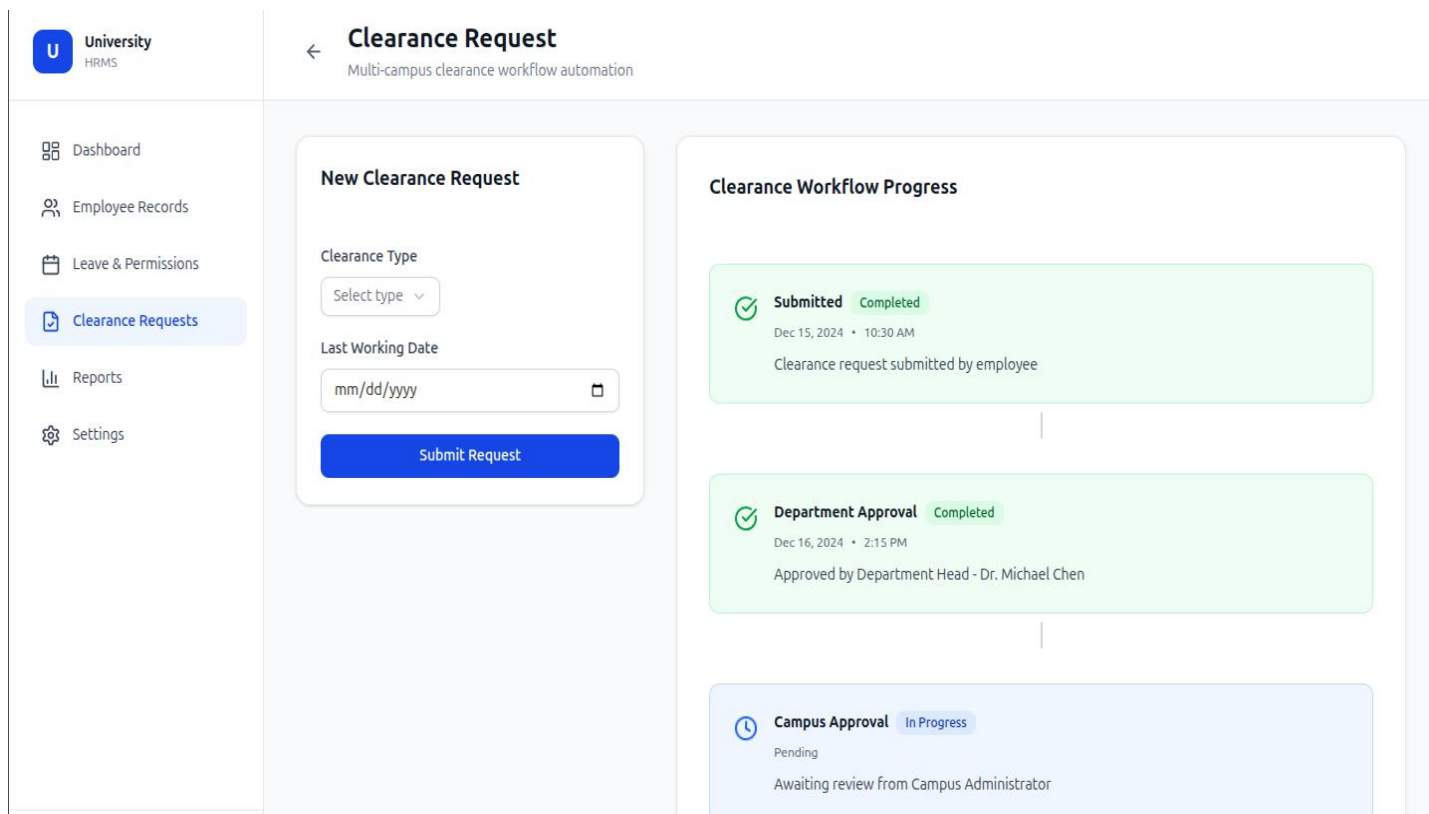


Figure 2.18 : Clearance request UI prototype

U

University
HRMS

Dashboard

Employee Records

Leave & Permissions

Clearance Requests

Reports

Settings

←

Employee Profile

View and manage employee information



Sarah Johnson

Senior LecturerActive

EMPLOYEE ID

EMP-2024-001

DEPARTMENT

Computer Science

CAMPUS

Main Campus

JOIN DATE

Jan 15, 2018

sarah.johnson@university.edu

+1 (555) 123-4567

Personal Information

Employment Details

Leave History

Clearance Status

DATE OF BIRTH

March 22, 1985

GENDER

Female

NATIONALITY

American

MARITAL STATUS

Married

ADDRESS

EMERGENCY CONTACT

Figure 2.19 :Employee profile UI prototype

Chapter 3: System Design

This chapter presents the system design of the proposed Human Resource Management System (HRMS) for Bahir Dar University. It translates the requirements outlined in the previous chapter into a structured and detailed blueprint for development. The design phase encompasses both high-level architectural strategy and low-level component specifications, including subsystem architecture, database structure, workflows, user interfaces, and system security mechanisms.

For the HRMS—which automates clearance management, leave processing, sabbatical eligibility verification, and payroll name transfers—system design emphasizes operational efficiency, consistent policy enforcement, and centralized data integrity across campuses and administrative units. The platform supports digital workflows, real-time status tracking, and institutional integration with Finance, while providing user-friendly interfaces for employees, supervisors, and HR officers. Security, performance optimization, and reliability are treated as core design considerations to ensure the platform effectively modernizes HR practices and reduces the administrative burden associated with manual paper-based procedures. This documentation serves as a guiding framework for the development team throughout the system lifecycle.

Design Goals

Design goals define the expected qualities and operational characteristics of the HRMS, reflecting non-functional requirements such as performance, security, reliability, scalability, and maintainability. These objectives ensure the system not only provides functional capabilities but also operates efficiently, consistently, and in alignment with institutional priorities.

Functionality

The system is designed to provide a comprehensive set of HR capabilities, including online submission and processing of clearance, leave, and sabbatical requests; automated eligibility verification for sabbatical leave; secure payroll data transfer to the Finance department; and centralized storage and retrieval of employee information. The design aims to reduce paper-based transactions, eliminate redundant administrative activities, improve transparency between departments, and increase accuracy in HR records. In doing so, the HRMS supports more efficient coordination across campuses and ensures consistent application of institutional policies.

Performance

Performance is a key design priority due to the multi-campus nature of the university and the high volume of HR transactions it manages. The system must provide rapid response times for user requests, real-time retrieval of employee data, and efficient workflow execution. Operations such as clearance status updates, leave balance calculations, and sabbatical eligibility checks are expected to be processed with minimal latency. The platform is also designed to support concurrent access by multiple administrative units, ensuring that performance remains stable as institutional workload scales.

Dependability

Dependability encompasses reliability, availability, security, and error handling. Reliability is supported through centralized database controls, consistent workflow execution, and mechanisms that preserve transaction integrity. High availability is ensured by designing the system to support continuous operation, allowing administrative activities to proceed without unnecessary delays or downtime. Security is enforced through authentication, authorization, secure communication, and audit logging, protecting both personal employee data and financial information involved in payroll coordination. Error-handling strategies include input validation, workflow exception handling, diagnostic logging, and structured feedback that informs users of incorrect submissions or rejected requests. These mechanisms collectively ensure stable and trustworthy operation.

Maintainability

Maintainability is addressed through comprehensive documentation, modular subsystem architecture, and structured error monitoring. Documentation supports future training, updates, and integration efforts. Modular workflows—separating clearance, leave, sabbatical, payroll, and authentication processes—allow enhancements to be introduced selectively without disrupting the entire system. Continuous monitoring of system logs and request transactions enables prompt identification of errors, improving operational resilience. Scalability is also incorporated into maintainability by allowing the system to accommodate future increases in user population, additional campuses, and evolving HR policies without substantial redesign.

End-User Criteria

End-user considerations play a crucial role in system design due to the diversity of actors involved, including employees, HR officers, supervisors, and financial administrators. The platform incorporates

an intuitive interface that simplifies navigation, form submission, record viewing, and status tracking, ensuring accessibility for users with varying levels of technical proficiency. By automating workflows that previously required physical document exchange, the system improves administrative efficiency and reduces the time required to approve clearances, leaves, and sabbatical requests. Transparency is achieved through real-time visibility of request status and digital records, eliminating information gaps and reducing the need for manual follow-up inquiries. The system further enhances convenience by minimizing the need for inter-campus travel or repeated office visits, aligning HR processes with modern expectations for digitized institutional services.

3.1 Architectural Design

The HR Management System (HRMS) is structured as a web-based application utilizing a three-tier architectural model. This architecture improves scalability, maintainability, and system performance by segmenting the application into three major layers:

1. Presentation Layer (Client-Side User Interface)

This layer manages all user interactions through a web or mobile interface. It provides dashboards, forms, approval views, and status notifications for employees, HR personnel, and management. Communication with the application layer occurs via HTTP/HTTPS requests and RESTful APIs.

2. Application Layer (Business Logic & Service Processing)

This layer handles client requests and executes core HR business logic such as:

- employee onboarding,
- leave approvals,
- payroll calculations,
- performance evaluations,
- attendance processing.

It mediates between the presentation and data layers by validating input, applying organizational rules, and enforcing security policies. It integrates with external systems including:

- payroll and banking services,
- authentication providers,

- reporting tools.

This layer ensures proper coordination of workflows across multiple HR subsystems.

3. Data Layer (Database Server & External Data Integrations)

This layer manages persistent data storage, retrieval, and modification for employee records, payroll histories, attendance logs, and performance evaluations. It includes:

- relational database servers,
- secure storage for sensitive employee data,
- compliance repositories for audit and reporting.

It supports external integration with financial and regulatory systems for payroll and benefits processing.

The data layer ensures consistency, data integrity, and secure access control.

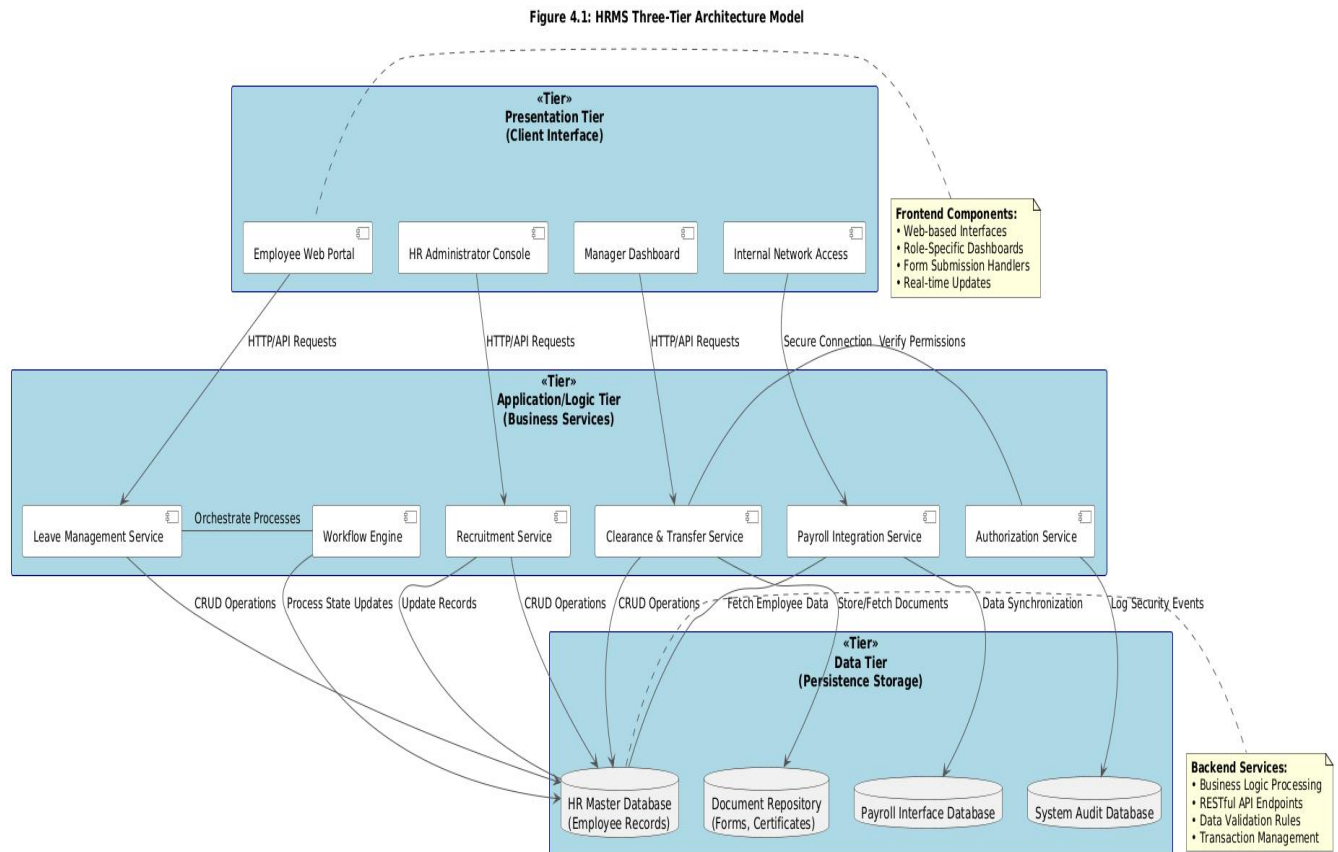


Figure 3.1: Three-tier architecture model

Subsystem

System decomposition is a fundamental aspect of system design, enabling the breakdown of a complex Human Resource Management System (HRMS) into smaller, more manageable subsystems. This structured approach enhances system comprehensibility, maintainability, and scalability by organizing related functionalities into distinct operational modules.

Each subsystem operates as a self-contained unit comprising classes, associations, operations, events, and constraints that collectively fulfill specialized HR responsibilities. These subsystems are integrated to support seamless interactions and ensure cohesive end-to-end process execution across the HR domain.

The subsystems of the HRMS are as follows:

- **Internal Job Posting and Vacancy Announcement Subsystem**

Manages job requisitions, internal vacancy postings, candidate routing, and competitive placement of existing employees. It facilitates internal mobility by advertising opportunities and processing applications within the organization.

- **Electronic Leave Management Subsystem**

Handles the full leave lifecycle including request submission, approval workflows, leave balance tracking, and notifications. It automates the process for employees and managers, ensuring policy compliance and real-time visibility.

- **Employee Clearance Management Subsystem**

Manages employee clearance activities during transfers, relocations, or exits, ensuring compliance with internal procedures and departmental approvals. It orchestrates checklists across departments (e.g., IT, Finance) to formalize the separation or transfer process.

- **Digital Employee Name Transfer for Payroll Processing Subsystem**

Automates and synchronizes employee name transfers and updates for payroll operations, reducing administrative errors and delays. This subsystem ensures that any formal name change or transfer of employment details is accurately reflected in the payroll system.

- **Centralized Data Management Subsystem**

Serves as the single source of truth, providing foundational data management for all

employee master records, personal information, and employment history. It is the core repository that feeds consistent and secure data to all other operational subsystems.

- **User Interface (Portal) Subsystem**

Provides role-specific, unified access to all HRMS functionalities through tailored dashboards and interfaces for employees, managers, and HR administrators, enabling intuitive interaction with the system.

Figure 3.2: HRMS Subsystem Model

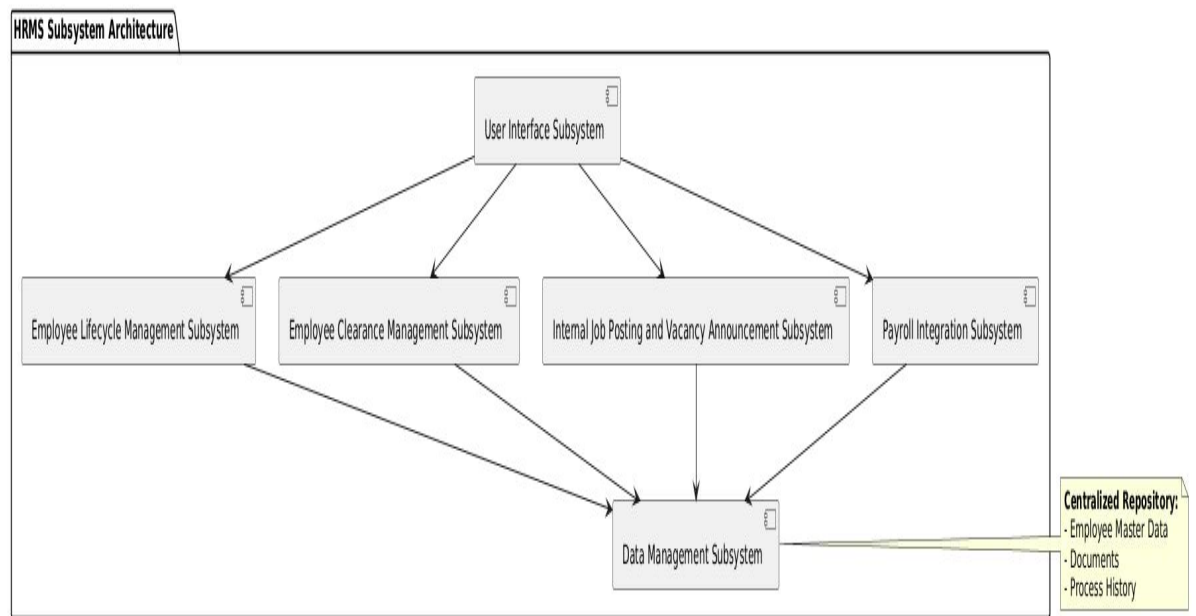


Figure 3.2: Subsystem model

3.1.1 Component Modeling

In a component-based architecture, the HRMS (Human Resource Management System) is decomposed into modular, self-contained units known as components. Each component encapsulates a specific HR function or service, contributing to the overall system in a structured and maintainable manner. These components are designed to be loosely coupled, enabling them to operate independently while supporting flexibility, scalability, and ease of maintenance.

Although each component functions autonomously, it collaborates with others through well-defined interfaces and service calls, ensuring seamless data flow and process integration across the system. This

architectural approach allows the HRMS to efficiently manage core HR tasks—such as employee clearance, electronic leave management, payroll i

ntegration, and internal recruitment—without creating tight dependencies between components.

HRMS System Component

The HRMS System component encompasses the primary operational modules that execute core human resource functions. It enables HR administrators, managers, and employees to interact with essential HR processes through dedicated interfaces.

The HRMS System includes the following components:

Employee Clearance Management: This component handles the end-to-end clearance process for employees during transfers, resignations, or exits, ensuring all departmental approvals are completed systematically.

Electronic Leave Management: This component manages the entire leave lifecycle—from application and approval workflows to balance tracking and notifications—providing a digital alternative to paper-based processes.

Digital Employee Name Transfer for Payroll: This component automates the synchronization of employee name updates and transfers with the payroll system, reducing manual errors and processing delays.

Internal Job Posting and Vacancy Announcement: This component facilitates the creation, publication, and management of internal job openings, enabling transparent internal mobility and talent placement.

Security Component

The Security component ensures system integrity and data protection through access control and encryption mechanisms. It includes:

Role-Based Access Control: This interface manages user permissions and authorization based on organizational roles, ensuring that employees, managers, and HR staff access only the functions relevant to their duties.

System Encryption: This interface secures sensitive data both at rest and in transit, safeguarding employee personal information and confidential HR documents.

Database Component

The Database component serves as the centralized repository for all HR-related data, supporting persistence, retrieval, and integrity of information. It consists of:

Centralized Employee Data: This interface maintains a single source of truth for employee records, including personal details, employment history, and role information.

Data Persistence Model: This interface defines how data is stored, indexed, and retrieved, ensuring reliability and performance for all HR transactions and reports

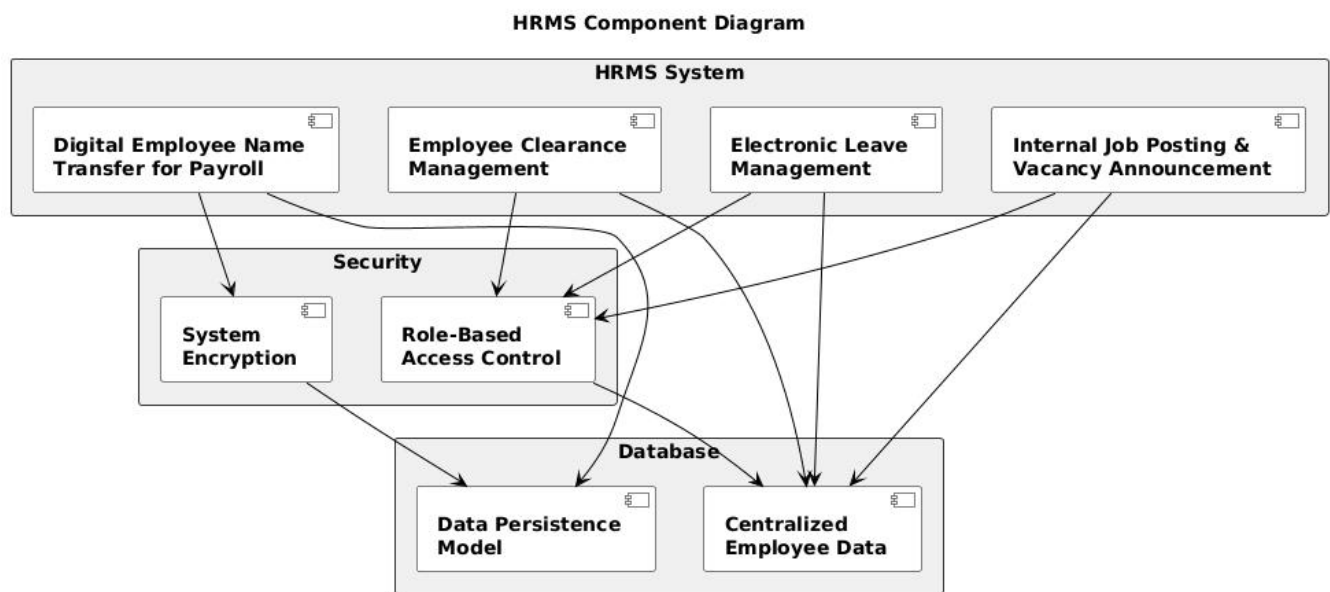


Figure 3.3: Component Diagram

3.2 Deployment Modeling

The HRMS (Human Resource Management System) is deployed using a modern cloud-based three-tier architecture model, leveraging specialized platforms for optimal performance, scalability, and cost-efficiency.

1. Client Layer (Presentation Layer)

Client Devices: University staff access via web browsers

Hosting: Vercel platform with global CDN distribution

Technologies: React.js, HTML5, CSS3, JavaScript

Security: HTTPS encryption for all communications

2. Application Layer (Business Logic & API)

Backend Hosting: Render platform

Runtime: Node.js/Express.js environment

API Architecture: RESTful API with JWT authentication

Features: Auto-scaling, environment management, health checks

3. Data Layer (Database)

Database Hosting: Render PostgreSQL

Features: Automated backups, connection pooling, read replicas

ORM: Prisma for database operations and migrations

Figure 3.4: HRMS Cloud Deployment Architecture

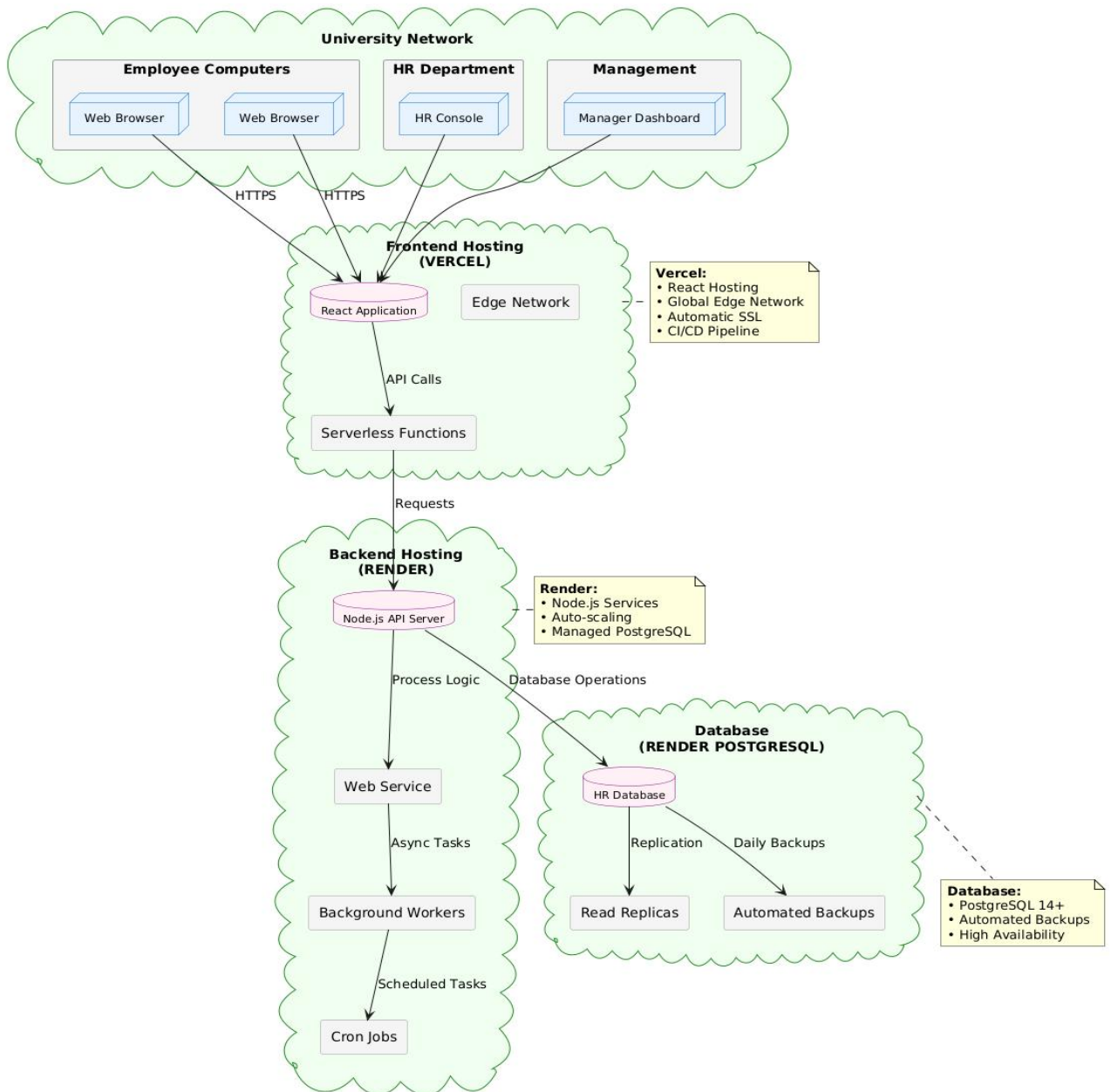


Figure 3.4: Deployment diagram

3.2.1 Design Class Model

The Design Class Model represents the object-oriented structure of the HRMS system, defining classes, attributes, methods, relationships, access visibility, and multiplicity. This model ensures a clear understanding of how different components interact within the system.

Each class in the system is described with:

Attributes: Data members, their data types, and access modifiers (public, private, protected).

Methods: Functions associated with the class, specifying return types and parameters with data types.

Relationships: Associations between classes (one-to-one, one-to-many, etc.), including inheritance and dependencies.

Multiplicity: Defines how many instances of one class relate to another (e.g., 1, 0.., 1..).

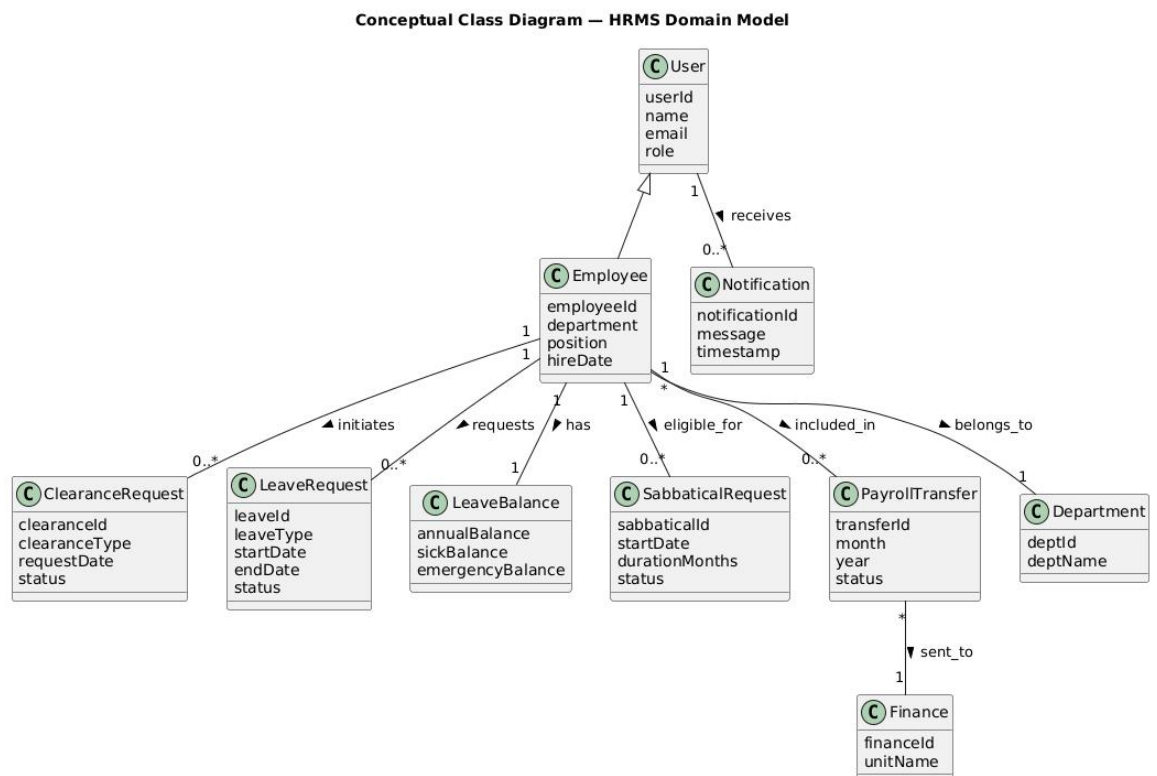


Figure 3.6: Class diagram

Entity Descriptions

Table 3.1: User Entity Description

Attribute Name	Data Type	Length	Constraint
userID	String	30	PRIMARY KEY
name	String	100	NOT NULL
email	String	100	NOT NULL, UNIQUE
password	String	255	NOT NULL
role	String	50	NOT NULL
status	String	20	DEFAULT 'Active'
createdAt	DateTime	-	NOT NULL
lastLogin	DateTime	-	-

Table 3.2: Employee Entity Description

Attribute Name	Data Type	Length	Constraint
employeeID	String	30	PRIMARY KEY
userID	String	30	FOREIGN KEY (User)
departmentID	String	30	FOREIGN KEY (Department)
position	String	100	NOT NULL
hireDate	Date	-	NOT NULL
employmentType	String	50	NOT NULL

Attribute Name	Data Type	Length	Constraint
salaryGrade	String	20	-
supervisorID	String	30	-

Table 3.3: LeaveRequest Entity Description

Attribute Name	Data Type	Length	Constraint
leaveID	String	30	PRIMARY KEY
employeeID	String	30	FOREIGN KEY (Employee)
leaveType	String	50	NOT NULL
startDate	Date	-	NOT NULL
endDate	Date	-	NOT NULL
status	String	20	DEFAULT 'Pending'
reason	String	500	-
approvedBy	String	30	-
approvalDate	DateTime	-	-

Table 3.4: ClearanceRequest Entity Description

Attribute Name	Data Type	Length	Constraint
clearanceID	String	30	PRIMARY KEY
employeeID	String	30	FOREIGN KEY (Employee)
clearanceType	String	50	NOT NULL

Attribute Name	Data Type	Length	Constraint
requestDate	DateTime	-	NOT NULL
status	String	20	DEFAULT 'Pending'
completionDate	DateTime	-	-
remarks	String	500	-

Table 3.5: SabbaticalRequest Entity Description

Attribute Name	Data Type	Length	Constraint
sabbaticalID	String	30	PRIMARY KEY
employeeID	String	30	FOREIGN KEY (Employee)
startDate	Date	-	NOT NULL
durationMonths	Integer	-	NOT NULL
status	String	20	DEFAULT 'Pending'
approvedBy	String	30	-
approvalDate	DateTime	-	-
reason	String	500	-

Table 3.6: PayrollTransfer Entity Description

Attribute Name	Data Type	Length	Constraint
transferID	String	30	PRIMARY KEY
employeeID	String	30	FOREIGN KEY (Employee)
month	Integer	-	NOT NULL
year	Integer	-	NOT NULL
status	String	20	DEFAULT 'Pending'
transferDate	DateTime	-	-
processedBy	String	30	-
remarks	String	500	-

Class Relationships Summary

User → Employee (1:1): Each User has exactly one Employee record

Employee → Department (N:1): Many Employees belong to one Department

Employee → LeaveRequest (1:N): One Employee can submit multiple LeaveRequests

Employee → ClearanceRequest (1:N): One Employee can initiate multiple ClearanceRequests

Employee → SabbaticalRequest (1:N): One Employee can request multiple Sabbatical leaves

Employee → PayrollTransfer (1:N): One Employee has multiple payroll transfer records

ClearanceRequest → DepartmentApproval (1:N): One ClearanceRequest requires approvals from multiple Departments

User → Notification (1:N): One User can receive multiple Notifications

3.2.2 Entity-Relationship (E-R) Model

The Entity-Relationship (E-R) Model represents the logical structure of the HRMS relational database. It defines how data is organized, stored, and related within the system.

The E-R Model consists of the following key elements:

Entities: Represent real-world objects (e.g., Employee, Department, Leave, Payroll).

Attributes: Define the properties of each entity, including data types and constraints.

Primary Keys (PKs): Unique identifiers for each entity.

Foreign Keys (FKs): Establish relationships between different entities.

Relationships: Define how entities are connected (one-to-one, one-to-many, many-to-many).

Multiplicity: Specifies the number of instances involved in a relationship (e.g., 1, 0.., 1..).

The diagram below illustrates the E-R Model for HRMS, showing the relationships between key entities and their attributes.

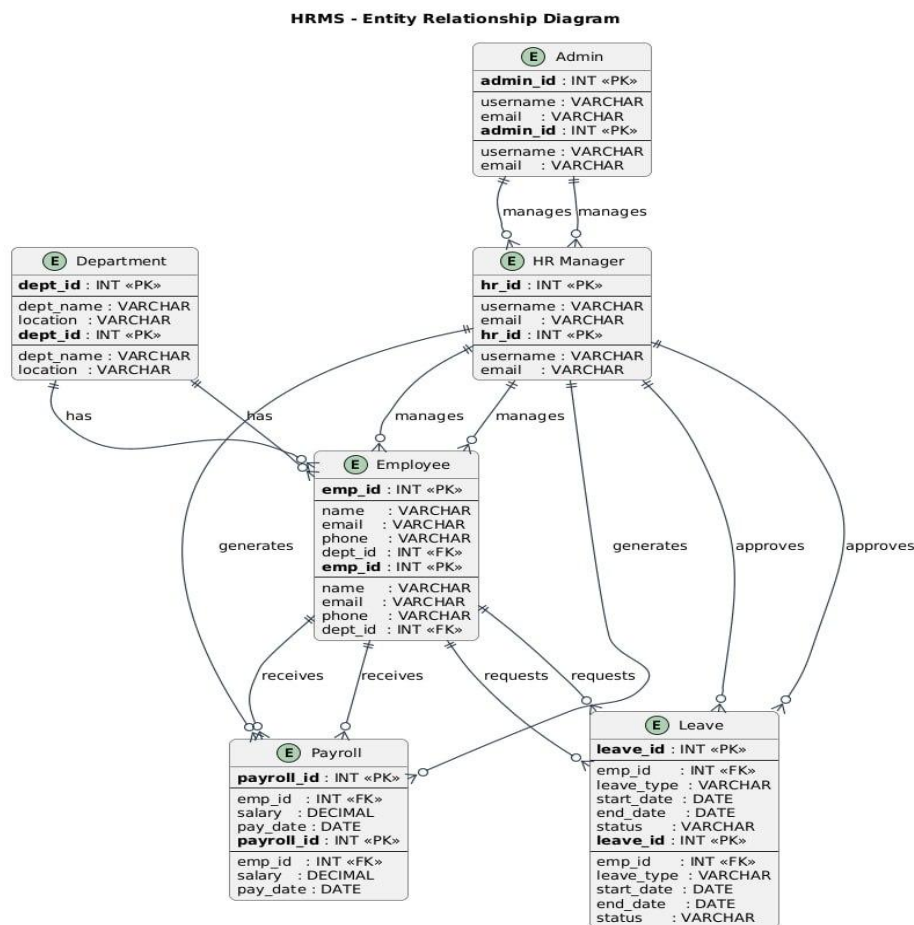


Figure 3.7: E-R diagram

U

University HRMS

Human Resource Management System

Employee ID

Enter your employee ID

Password

Enter your password

Forgot password?

Login

For support, contact HR at hr@university.edu

Figure 3.8 : login prototype UI

U

University
HRMS

Dashboard

Employee Records

Leave & Permissions

Clearance Requests

Reports

Settings

Dashboard

Welcome back, Admin

Total Employees

1,247

Pending Registrations

23

Pending Leave Requests

45

Pending Clearance Requests

12

Recent Requests

Latest submissions requiring attention

View All

TYPE	EMPLOYEE	DETAILS	DATE	STATUS	ACTION
Leave Request	Sarah Johnson EMP-2024-001	Annual Leave	Dec 15, 2024	Pending	Review
Clearance Request	Michael Chen EMP-2024-002	Resignation	Dec 14, 2024	Approved	Review
Leave Request	Emma Davis EMP-2024-003	Sick Leave	Dec 13, 2024	Rejected	Review
Registration	James Wilson EMP-2024-004	New Employee	Dec 12, 2024	Pending	Review
Clearance Request	Lisa Anderson EMP-2024-005	Retirement	Dec 11, 2024	Approved	Review

Figure 3.9 :Hr Admin Dashboard UI prototype

U

University
HRMS

←

Clearance Request

Multi-campus clearance workflow automation

Dashboard

Employee Records

Leave & Permissions

Clearance Requests

Reports

Settings

New Clearance Request

Clearance Type

Select type ▾

Last Working Date

mm/dd/yyyy

Submit Request

Clearance Workflow Progress

✓

Submitted

Completed

Dec 15, 2024 • 10:30 AM

Clearance request submitted by employee

✓

Department Approval

Completed

Dec 16, 2024 • 2:15 PM

Approved by Department Head - Dr. Michael Chen

🕒

Campus Approval

In Progress

Pending

Awaiting review from Campus Administrator

Figure 3.10 : Clearance request UI prototype

3.4 Access Control and Security

In our Human Resource Management System (HRMS), access control and security are implemented to ensure that only authorized users can access system functionalities and sensitive data. The system follows a role-based access control (RBAC) approach, where permissions are granted based on the user's assigned role. This helps maintain data confidentiality, integrity, and proper system usage.

Table 3.7: Access Control and Security

Task	Admin	HR Manager	Employee
System Management			
Manage users & roles	✓		
System configuration	✓		

Task	Admin	HR Manager	Employee
Employee Records			
View all employees	✓	✓	
Add/update employees		✓	✓ (own)
Leave Management			
Approve leave		✓	
Apply for leave			✓
Payroll			
Generate payroll		✓	
View payroll	✓	✓	✓ (own)
Clearance			
Process clearance		✓	
Request clearance			✓
Job Postings			
Create postings		✓	
View vacancies	✓	✓	✓
Login & Access	✓	✓	✓

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